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**Cite this article:** Kumar K R, Dutta PS. 2026Rate-induced tipping in savanna–forest ecosystems: effects of fast-varying fitness parameters in the framework of compactification. *Proc. R. Soc. A* **482**: 20250803.<https://doi.org/10.1098/rspa.2025.0803>

Received: 16 September 2025

Accepted: 16 March 2026

Subject Areas:

applied mathematics, mathematical modelling

Keywords:

non-autonomous systems, R-tipping, savanna–forest ecosystems, basin instability, time-dependent external input, compactification

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e-mail: parthasharathi@iitrpr.ac.inElectronic supplementary material is available online at <https://doi.org/10.6084/m9.figshare.c.8407476>.**THE ROYAL SOCIETY**
PUBLISHING

Rate-induced tipping in savanna–forest ecosystems: effects of fast-varying fitness parameters in the framework of compactification

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Savannas are grass-dominated ecosystems with scattered shrubs and savanna trees, often coexisting with tropical forests. They host a diverse range of plants and animals, act as a major carbon sink, regulate the global climate and support local economies. However, rapid environmental change is degrading these ecosystems, ultimately leading to a decline in biodiversity. Here, we consider a mathematical model of savanna–forest ecosystems to investigate rate-induced tipping (R-tipping) between distinct states under rapid variations in fitness parameters of different functional types (i.e. grass, savanna saplings and adult trees and forest trees) over time. To establish the existence of R-tipping, we determine basin instability (BI) in the corresponding frozen system with fixed-in-time inputs. Since time-dependent parameters make the model non-autonomous, it lacks compact invariant sets, which preclude the use of classical bifurcation theory. To address this, we reformulate the non-autonomous system with bi-asymptotically constant inputs as an autonomous one using compactification with an additional bounded variable. Furthermore, compactification enables us to identify the R-tipping threshold, which is crossed when tipping occurs. Overall, our analysis reveals distinct rate-induced transitions, including shifts to dominant states of grass, adult savanna trees and oscillatory dynamics.

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1. Introduction

Ecosystems, which bring together living and non-living things, are complex dynamic systems that continuously evolve and play a vital role in sustaining life on Earth. However, over the past few decades, many ecosystems have been deteriorating rapidly owing to multiple stressors. They risk collapsing from a favourable state for living organisms to an unfavourable one, primarily due to climate change and human activities. For example, the ongoing degradation of the Amazon forest ecosystem is driven by stressors such as rising temperatures, extreme droughts, deforestation and forest fires [1,2]. Recent assessments by the Intergovernmental Panel on Climate Change (IPCC) [3] and the Global Tipping Points Report 2025 [4] suggest that, under such continued stressors, the Amazon forest may face an elevated risk of collapsing before the end of the 21st century. Similarly, the coral reefs are transitioning to the state of macroalgal dominance driven by bleaching, overfishing and ocean acidification [5], along with many other ecosystems facing similar threats [6–10].

Specifically, these ecosystems may undergo sudden, nonlinear and often irreversible transitions to a contrasting state, known as critical transitions or tipping events [11]. From a dynamical systems perspective, such transitions are often associated with crossing critical levels or dangerous bifurcation points [12] in response to gradual or rapid changes in external conditions. This phenomenon is referred to as bifurcation-induced tipping [13,14]. However, under current unprecedented environmental changes, ecosystems may shift to an alternative stable state even before reaching the predicted critical levels if the rate of change in external conditions exceeds a specific critical rate [15–17]. Such a tipping mechanism can be understood using non-autonomous dynamical systems theory with time-varying external conditions and is known as rate-induced tipping (R-tipping) [11,18–20]. R-tipping occurs when the rate of change in an external input exceeds a critical rate, causing the system to deviate from its moving base state (attractor) and cross the basin boundary, tipping to an alternative state [19,21–26]. The existence of R-tipping highlights that ecosystem stability depends not only on the magnitude of environmental change but also on its rate, complicating prediction and control [27]. Therefore, identifying and analysing ecosystems that exhibit such tipping behaviour is paramount.

This work examines how the rate of change in external conditions leads to rate-induced critical transitions between alternative stable states in savanna–forest ecosystems [28,29]. Savannas¹ are natural habitats for a wide variety of plants and animals, with some areas having up to 60 plants per square metre [30]. They span almost 20% of the Earth's landscape across tropical and subtropical regions and feature the coexistence of grasses, shrubs and scattered trees [31]. Diverse savannas can be found on different continents and the distribution of grasses and trees varies among them based on a variety of factors, including fire, soil fertility, grazing pressure, human activity and climatic conditions like drought and rainfall [32–35]. Beyond their ecological diversity, savannas play a crucial role in the global carbon cycle by capturing and storing large amounts of carbon, thereby regulating global climate [31]. Because global conservation and climate policies have historically prioritized tree cover and forests, savannas, being open, grass-dominated ecosystems with sparse tree cover, have often received less attention and recognition from policymakers [36]. In many regions, it was misclassified as a degraded forest [37]. As a result, these open, sparse lands are targeted for large-scale tree plantations that aim to increase the forest cover worldwide to mitigate global climate change [38,39]. Although savannas and tropical forest trees have long been in natural competition, acting as alternative stable ecosystem states [40,41], the current ‘forest invasion’ in the form of afforestation [42] is happening at a rapid pace, often ignoring the savannas’ distinctive ecological balance and biodiversity [39].

Fire plays a significant role in shaping the structure and distribution of savannas [43–48]. Fire creates positive feedback in maintaining savannas, where the forest is also a viable, stable ecosystem state, by controlling the invasion of forest trees and creating an open tree canopy, which results in high grass productivity and further fuels frequent fires [49,50]. Thus, fire suppression

¹In general, savannas correspond to grass-dominated systems with savanna trees present, including both saplings and adult savanna trees, but no tropical forest trees.

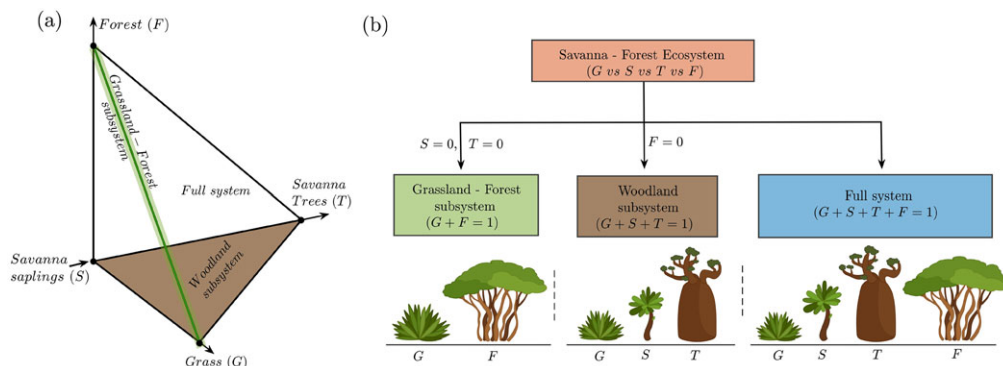


Figure 1. (a) A tetrahedral phase space (3-simplex) illustrating distinct dynamics of the savanna–forest system (2.1) with functional types such as grass (G), savanna saplings (S), adult savanna trees (T) and forest trees (F). The green-shaded line segment, constrained by $G + F = 1$, corresponds to the grassland–forest subsystem without savanna saplings ($S = 0$) and adult savanna trees ($T = 0$). The brown-shaded base triangle, subject to $G + S + T = 1$, represents the grassland–woodland subsystem (or simply woodland subsystem) with no forest trees ($F = 0$). Finally, the full system, incorporating all the components, is governed by the constraint $G + S + T + F = 1$. (b) A schematic flowchart outlining different subsystems in the savanna–forest system described in (a).

via any form (e.g. converting the savanna biome into agricultural lands or intense grazing) [51] can push the system to an alternative state, such as woody encroachment by savanna trees or the establishment and expansion of fire-sensitive forest tree species, leading to forest dominance [52]. On the other hand, frequent fires promote grass dominance by reducing woody biomass and limiting long-term carbon storage in trees. While grasslands retain significant amounts of underground carbon stored in their roots, repeated burning releases above-ground carbon and alters the ecosystem’s carbon dynamics [53]. Considering these dynamics, we study a global savanna–forest ecosystem model [35,54], which is autonomous, and incorporates time-varying fitness input parameters. These time-varying parameters are modelled as sudden pulses with narrow and broad peaks [55–57], representing ecosystem disturbances (for example, sudden forest fires [58], rapid increase in atmospheric carbon dioxide concentrations [16,20], climate warming [27], etc.) or benefits influenced by rate parameters. The autonomous model captures the proportions of grass (G), savanna saplings (S), adult savanna trees (T) and forest trees (F) under the key assumption that the total area covered by them sums to a constant. As a result, the system’s phase space is constrained within n -simplexes, where n represents the dimension of the system after applying the constraint. In addition, the system does not include an explicit parameter for fire. Instead, fire spread is determined by the amount of grass cover, thereby restricting the spatial component of fire spread and resulting in a nonlinear ordinary differential equation system. Depending on the presence or absence of certain functional types, the savanna–forest system can be divided into three subsystems with the constraint: (i) the grassland–forest subsystem, (ii) the grassland–woodland subsystem (or simply woodland subsystem) and (iii) the full system, where all components interact. The phase spaces of these subsystems vary from a line segment (1-simplex) to a tetrahedron (3-simplex) (see figure 1), with dynamics ranging from steady states to heteroclinic cycles [54].

R-tipping, being a non-autonomous instability, cannot be detected directly using classical bifurcation theory. The existence of R-tipping is instead guaranteed through an autonomous property known as basin instability (BI). The BI of an attractor is a global property of the corresponding autonomous frozen system with the fixed-in-time input parameters [26,59–61]. It serves as a key condition for a suitable external forcing, which can be used to analyse R-tipping in higher-dimensional systems. Once the input parameter turns time-dependent, the non-autonomous systems lack compact invariant sets such as equilibria, limit cycles or chaotic

attractors to observe tipping between them. To overcome this, we utilize bi-asymptotically constant time-varying external inputs [62,63] to access the attractors from both past and future limit systems as time tends to $-\infty$ and $+\infty$. Here, in the savanna–forest system, we employ two distinct non-monotone time-dependent inputs that act on the growth, maturation and fire-induced mortality parameters of different functional types to capture rate-induced transitions.

Although the savanna–forest model approaches an autonomous limit as the bi-asymptotic external inputs converge to constants, the system remains non-autonomous at all finite times and only becomes autonomous asymptotically. To incorporate these asymptotic limit systems into the analysis, we compactify the phase space by augmenting a bounded but open dimension and extending it at both ends with flow-invariant subspaces that encapsulate the autonomous dynamics of the limit systems [26,64,65]. The resulting augmented system is fully autonomous, and its additional eigendirections constitute invariant manifolds that approximate the moving stable equilibrium in the non-autonomous savanna–forest system. This framework allows us to examine R-tipping in a rigorous geometric setting. Further, implementing this compactification approach for the three subsystems reveals distinct routes to R-tipping. In the grassland–forest subsystem, transitions lead to grass dominance, with tipping occurring from a low grass cover equilibrium in the past limit system to a high grass cover equilibrium in the future limit system upon varying birth and high fire-induced mortality rates of forest trees following both inputs set up in time. In the compactified woodland 2-simplex subsystem, we observe the rate-induced regime shift from low to high savanna forest tree cover on varying recruitment and maturation rates of savanna trees. Finally, the compactified full system transitions from the coexistence equilibrium to an oscillatory attractor upon varying the recruitment rates of savanna and forest trees, respectively.

The paper is organized as follows: §2 describes the savanna–forest mathematical model and different subsystems. The methods §3 explains the concepts of BI, time-varying external inputs and compactified problem setting for R-tipping analysis. The results section (§4) is divided into three subsections, each examining the concept of BI in the autonomous frozen subsystems and rate-induced critical transitions through tipping diagrams. Section 4a analyses the grassland–forest subsystem, highlighting grass-driven dynamics over forest trees. Section 4b focuses on the grassland–woodland subsystem, where R-tipping leads to a transition from low to high savanna forest cover. In §4c, the full system is explored, detailing the transition from a stable steady state to a periodic solution. Finally, §5 concludes the study.

2. Mathematical model of savanna–forest ecosystems

We consider a stage-structured nonlinear model of savanna–forest ecosystems [54] that accounts for the interplay between grasses (G), savanna saplings (S), adult savanna trees (T) and forest trees (F). A key assumption is that the entire space is occupied by either one or more of these functional types, ensuring the proportions of their land cover sum to a constant [35], thereby imposing a constraint on the model. The global ecological savanna–forest autonomous system, subject to the constraint $G + S + T + F = 1$, is described by the following system of equations:²

$$\left. \begin{aligned} \dot{G} &= \mu S + \nu T - \beta GT + \varphi(G, S, T)F - \alpha GF, \\ \dot{S} &= \beta GT - (\omega(G, S, T) + \mu)S - \alpha SF, \\ \dot{T} &= \omega(G, S, T)S - \nu T - \alpha TF \\ \text{and} \quad \dot{F} &= \alpha F(1 - F) - \varphi(G, S, T)F, \end{aligned} \right\} \quad (2.1)$$

where μ and ν represent the mortality rates of savanna saplings and adult savanna trees, respectively, contributing to a positive feedback mechanism supporting grass abundance. The parameters β and α denote the recruitment rates of savanna saplings and forest trees, respectively. The model is primarily shaped by a fire–vegetation feedback mechanism, grounded in two

²The dot ‘.’ on the top of the variables refers to a derivative with respect to time.

key ecological observations. The first is that the fire limits the emergence of large savanna trees by affecting the maturation rate of savanna saplings to adult trees. While saplings are highly fire-susceptible, mature savanna trees are largely resistant to moderate fires owing to traits such as thick bark, high moisture content and open canopy structures [49,66]. This size-dependent fire response motivates the stage-structured representation of saplings (S) and adults (T) in the model. In contrast, forest trees (F) are highly fire-sensitive across life stages, so a single functional type is considered to capture their fire-driven dynamics. Examples of fire-resistant savanna trees include umbrella thorn acacia, African baobab and Australian eucalyptus. The second observation highlights that tree cover can suppress fire spread, especially when the density exceeds a critical threshold (approx. 40%) [34,67]. Although fire spreads spatially, the model does not explicitly include a spatial component [48], thereby restricting it to an ordinary differential equation. Still, a fire spread is modelled using mean-field approximations based on the idea that it is a percolation process [68,69]. Thus, the model does not have an explicit fire parameter; instead, the maturation rate from savanna saplings to trees depends on the amount of grass biomass. For instance, there is a lower chance of fire spreading in areas with low grass cover, implying that saplings can survive the fire, resulting in a higher maturation rate. On the other hand, with high grass cover, the maturation rate is low. In addition, the parameter $\gamma \in [0, 1]$ quantifies the extent to which savanna saplings (S) and adult savanna trees (T) permit fire spread by allowing grass to persist beneath their canopy, with $\gamma = 1$ corresponding to no suppression of fire spread relative to grass-only areas [54]. Consequently, the savanna sapling-to-adult tree maturation rate ω is expressed as a decreasing sigmoidal function depending on G, S and T , and it is given by

$$\omega(G, S, T) = \omega_0 + \frac{(\omega_1 - \omega_0)}{1 + e^{-(G+\gamma(S+T)-\theta_1)/s_1}}, \quad (2.2)$$

where ω_0 and ω_1 represent, respectively, maximum and minimum maturation rates between which the sigmoidal function is varied; θ_1 is a half-response threshold of the effective grass and savanna cover $G + \gamma(S + T)$ at which the maturation rate transitions between ω_0 and ω_1 , while s_1 controls the steepness of the transition.

Now, in the case of forest trees (F), which differ from adult savanna trees (T), growth is modelled using a logistic term involving the recruitment rate α . Forest tree mortality φ rate is fire-driven and increases with the effective grass and savanna cover that promotes fire spread. Low grass cover suppresses fire spread and results in lower forest tree mortality φ_0 , whereas high grass cover facilitates fire spread, to which forest trees are highly susceptible φ_1 . As in the savanna sapling-to-adult tree maturation term $\omega(G, S, T)$ equation (2.2), here, the parameter γ controls the degree to which savanna (S and T) cover permits fire spread by influencing grass cover. The mortality rate φ can be modelled with a sigmoidal function, and it is given by

$$\varphi(G, S, T) = \varphi_0 + \frac{(\varphi_1 - \varphi_0)}{1 + e^{-(G+\gamma(S+T)-\theta_2)/s_2}}, \quad (2.3)$$

where φ_1 and φ_0 denote the maximum and minimum mortality rates owing to fire spread under the influence of G, S and T . The parameters θ_2 and s_2 represent the half-response threshold and steepness of the sigmoidal function, respectively.

System (2.1) can be simplified into three subsystems under specific assumptions: (i) the absence of savanna saplings ($S = 0$) and adult savanna trees ($T = 0$), (ii) the absence of forest trees ($F = 0$) and (iii) the inclusion of all functional types. The first subsystem, the grassland-forest subsystem, encompasses the dynamics of grasses and forest trees and is constrained by the condition $G + F = 1$. The phase space reduces to a 1-simplex or a line segment with unit length, as indicated by the green shaded line in figure 1a. The second subsystem describes the interplay between grass, savanna saplings and adult savanna trees with the phase space reduced to a 2-simplex or a triangle with the condition $G + S + T = 1$ (brown shaded triangle in figure 1a). It is known as the grassland-woodland (or simply woodland) subsystem. Here, the term ‘woodland’ refers to a forest of adult savanna trees. The third case, or the full system, includes the dynamics of all functional types within a 3-simplex or a tetrahedral phase space. The

flow chart figure 1b describes all three subsystems of the original savanna–forest system (2.1). The following sections examine these subsystems and demonstrate rate-induced critical transitions in each corresponding non-autonomous subsystem by varying their fitness parameters over time, as listed in the table 1 in §S0 of electronic supplementary material (SI).

3. Methods

This section outlines the general framework from [26] for analysing R-tipping in the complete savanna–forest system (2.1) with time-dependent input parameters, which renders the system non-autonomous. To begin with, we consider a general non-autonomous system:

$$\dot{X} = f(X, \Gamma(rt)), \quad (3.1)$$

where the state variable $X \in \mathbb{S}^n \subset \mathbb{R}^{n+1}$ depends on the independent variable $t \in \mathbb{R}$ and we let $\mathbb{S}^n$ be the standard n -simplex, defined as follows:

$$\mathbb{S}^n = \left\{ (x_0, x_1, \dots, x_n) \in \mathbb{R}^{n+1} \left| \sum_{i=0}^n x_i = 1 \text{ and } x_i \geq 0 \text{ for } i = 0, 1, \dots, n \right. \right\}.$$

A nonlinear C^1 -smooth function $f: \mathbb{S}^n \times P \rightarrow \mathbb{S}^n$ governs the dynamics of the above system and we let $\Gamma: \mathbb{R} \rightarrow P$ be a C^1 -smooth non-autonomous external input with a rate parameter $r > 0$, where $P \subseteq \mathbb{R}$ represents an open parameter space³ of the corresponding autonomous frozen system:

$$\dot{X} = f(X, \Gamma), \quad (3.2)$$

which is traced by the time-varying external input $\Gamma(rt)$.

(a) Criterion leading to rate-induced tipping

Before investigating R-tipping, we first establish that the autonomous system (3.2) satisfies the conditions for the existence of R-tipping. Although autonomous bifurcation theory cannot be directly used to analyse R-tipping, certain conditions derived from the autonomous frozen system with fixed-in-time external inputs can predict R-tipping. One such criterion is ‘BI’, which was first introduced for exponentially stable equilibria in [60]. BI is a global property of the autonomous frozen systems involving the whole phase space with basins of attraction of the asymptotically stable attractors. To explore BI, the frozen system (3.2) must exhibit at least bistability with basin boundaries separating the basins of attraction of coexisting stable base equilibria (attractors in the frozen system) and a continuous parameter path (range) Δ_Γ without any bifurcation points. The basin boundaries can be the stable manifolds of a saddle equilibrium or an unstable limit cycle, and others. More formally, consider the frozen system (3.2) with a continuous parameter path Δ_Γ , and let $\{a(\Gamma): \Gamma \in \Delta_\Gamma\}$ be a C^1 -smooth branch of stable equilibria. For each $\Gamma \in \Delta_\Gamma$, let

$$B(a, \Gamma) = \left\{ X_0 \in \mathbb{S}^n : \lim_{t \rightarrow +\infty} X(t; X_0, \Gamma) = a(\Gamma) \right\}$$

denote the basin of attraction of $a(\Gamma)$, where $X(t; X_0, \Gamma)$ is the flow of the frozen system at parameter Γ . The stable equilibrium a is said to be basin unstable on Δ_Γ if there exists $\Gamma_1, \Gamma_2 \in \Delta_\Gamma$ such that

$$a(\Gamma_1) \in \mathbb{S}^n \setminus \overline{B(a, \Gamma_2)},$$

where $\overline{B(a, \Gamma)}$ is the closure of $B(a, \Gamma)$. Equivalently, BI occurs when the mapping

$$(\Gamma_1, \Gamma_2) \mapsto a(\Gamma_1) \in B(a, \Gamma_2)$$

fails for some $(\Gamma_1, \Gamma_2) \in \Delta_\Gamma \times \Delta_\Gamma$. The set of all parameter points $\Gamma_2 \in \Delta_\Gamma$ for which the equilibrium $a(\Gamma_1)$ is basin unstable constitutes the region of BI.

³In general, the parameter space can be any d -dimensional subspace of $\mathbb{R}^n$.

In [26], the authors proposed that the BI serves as a sufficient condition for the occurrence of R-tipping in the multi-dimensional system. Specifically, if a base attractor in the corresponding frozen system is basin unstable along the chosen parameter path, then a suitable external forcing exists that traces out the parameter path and induces R-tipping from that attractor to another attractor in the non-autonomous system. Notably, BI is a sufficient, but not a necessary, condition to guarantee R-tipping. One can find examples where R-tipping between attractors occurs without the BI in [70]. Once BI is observed, the next step is to determine the suitable external forcing that results in R-tipping when the rate of forcing exceeds a critical rate. In continuation with this, the following section explains the two identified external forcing of different forms that lead to R-tipping in the savanna–forest system.

(b) Time-dependent external inputs

The non-autonomous system (3.1) can be transformed into an autonomous extended system by augmenting an additional dependent variable $s = rt$. The autonomous extended system defined on the phase space $S^n \times \mathbb{R}$ is given by

$$\left. \begin{aligned} \dot{X} &= f(X, \Gamma(s)) \\ \dot{s} &= r, \end{aligned} \right\} \quad (3.3)$$

where s is the added state variable. This makes it evident that the non-autonomous system (3.1) does not possess any compact invariant sets such as equilibria, limit cycles or tori, making it inaccessible to use autonomous theory. The primary cause of this disadvantage is the unbounded nature of the additional variable $s = rt$. To overcome this, we employ a general compactification framework,⁴ which incorporates bi-asymptotically constant external inputs as t tends to positive and negative infinity in the extended system, ensuring that the variable s remains bounded [65]. The resulting compactified autonomous system enables us to use the autonomous theory to analyse the behaviour of the original non-autonomous system (3.1). Specifically, we say the external input $\Gamma(rt)$ is bi-asymptotically constant with future limit Γ^+ and past limit Γ^- if

$$\lim_{t \rightarrow \pm\infty} \Gamma(rt) = \Gamma^\pm \in \mathbb{R}.$$

In particular, we consider two distinct non-monotone bi-asymptotic external inputs with fixed peaks, which are as follows:

$$\Gamma_1(rt) = \Delta_0 + \frac{\Delta_1}{2} \operatorname{sech}(r(t - t_c)) + \Delta_2 \tanh(r(t - t_c)), \quad (3.4)$$

with $\Gamma_1^\pm = \Delta_0 \pm \Delta_2 \in \mathbb{R}$, and

$$\Gamma_2(rt) = \delta_0 + \delta_1 [\tanh(r(t - t_{c1})) - \tanh(r(t - t_{c2}))], \quad (3.5)$$

with $\Gamma_2^\pm = \delta_0 \in \mathbb{R}$. The parameters Δ_i ($i = 0, 1, 2$) and δ_j ($j = 0, 1$) determine the input magnitudes. The input Γ_1 has a fixed peak at t_c , while the other input Γ_2 peaks between t_{c1} and t_{c2} for larger rate values. Although both inputs exhibit non-monotonicity and bi-asymptotic behaviour, they differ in other unique features. As $t \rightarrow \pm\infty$, the input Γ_1 asymptotically approaches different values, whereas Γ_2 returns to the same value. For given magnitudes, the rate parameter r influences only the steepness of the input Γ_1 , leaving its peak height unchanged at $\Delta_{\max}$. In contrast, the steepness and peak height of Γ_2 vary with r for the fixed magnitudes. For lower rates, the maximum value

⁴A well-established way to study non-autonomous systems is via the skew-product formalism, which treats them as autonomous flows in an augmented phase space [71]. Under the strong assumption of asymptotically autonomous dynamics, our compactification can be regarded as a special type of skew-product flow.

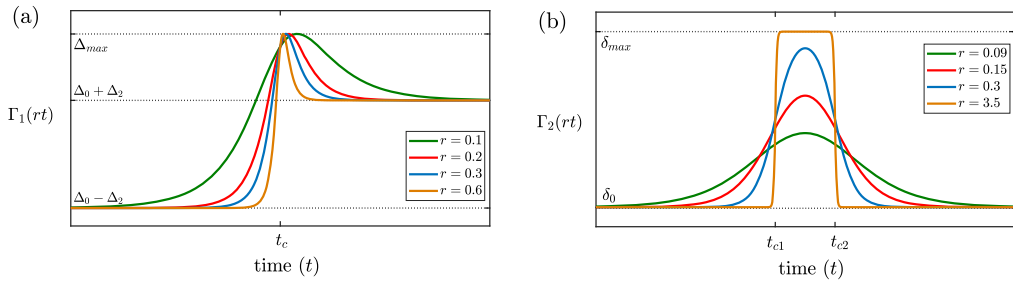


Figure 2. Examples of two sets of bi-asymptotically constant external inputs (a) equation (3.4) and (b) equation (3.5) varied over time.

of the input Γ_2 is

$$\Gamma_{2\max} = \delta_0 + 2\delta_1 \tanh\left(\frac{r(t_{c2} - t_{c1})}{2}\right),$$

which occurs at $t = \frac{t_{c1} + t_{c2}}{2}$ for the fixed magnitudes δ_j , $j = 0, 1$. For a sufficiently large r , the maximum approaches a constant value $\delta_{\max}$. Figure 2a,b illustrates these properties for the inputs Γ_1 and Γ_2 , respectively, with distinct rate values. As the rate increases from $r = 0.1$ to 0.6 in figure 2a, the steepness of the input Γ_1 increases, and the curves become narrower. The input Γ_2 in figure 2b changes both its width and peak height as the rate varies from $r = 0.09$ to 3.5 .

While R-tipping may also occur with monotone inputs (not necessarily always), our focus here is on non-monotone inputs. These inputs allow us to capture sudden pulses and peaked responses that settle to either the same or different constants as $t \rightarrow \pm\infty$ [61,62]. We choose this approach because savanna–forest ecosystems are regularly exposed to extreme disturbances such as intense fires, drought pulses and heavy wet-season storms, which typically reach a sudden peak over a short window [57,67], and as a result, the post-disturbance conditions can be either different or the same. The inputs Γ_1 and Γ_2 are designed to reflect these events, with Γ_1 fixing the peak height and Γ_2 allowing both the peak and width to vary with the rate, thereby highlighting how both the intensity and timing of disturbances can drive rate-induced tipping. Precise functional forms of these inputs are not intended to be predictive, but rather to represent a broad class of transient disturbances characterized by rapid onset and relaxation.

(c) Compactified system

The bi-asymptotic constant inputs Γ_1 and Γ_2 turn the non-autonomous system (3.1) asymptotically autonomous in the long run, that is,

$$f(X, \Gamma(rt)) \rightarrow f(X, \Gamma^\pm) \quad \text{as } t \rightarrow \pm\infty,$$

making the possibility of using the compact invariant sets of the respective autonomous future and past limit systems:

$$\dot{X} = f(X, \Gamma^\pm). \quad (3.6)$$

However, the dynamics of system (3.1) (and system (3.3)) do not contain these future and past limit systems as they only appear as t tends to positive and negative infinity. To overcome this, we augment the non-autonomous system (3.1) with an additional variable s defined by a parameterized coordinate transformation $s = g_\varepsilon(rt)$, where $g_\varepsilon : \mathbb{R} \rightarrow (0, 1)$. This transformation bounds s within an open interval, instead of leaving it unbounded as in the extended autonomous system (3.3). The system is then closed by attaching the limit system (3.6) from $t \rightarrow \mp\infty$ at $s = 0$ and $s = 1$, respectively. With these steps, the compactification ensures that the augmented system is continuously differentiable at the boundary points $s = 0$ and $s = 1$ [65].

To implement this, we use the exponentially decaying sigmoid function, consistent with the assumptions in appendix A(a), as the coordinate transformation. The compactification parameter $\varepsilon > 0$ quantifies the decay rate of the derivative of $g_\varepsilon(rt)$. The sigmoid function is chosen because it naturally maps the added variable s from $\mathbb{R}$ to $(0, 1)$, which aligns with the domain $\mathbb{S}^n$ after gluing $s = 0$ and $s = 1$ with $(0, 1)$. The coordinate transformation used is

$$s = g_\varepsilon(rt) = \frac{1}{1 + e^{-\varepsilon rt}}, \quad (3.7)$$

and its inverse function:

$$rt = h_\varepsilon(s) := g_\varepsilon^{-1}(s) = \frac{1}{\varepsilon} \ln \left(\frac{s}{1-s} \right). \quad (3.8)$$

Thus, the given external inputs equations (3.4) and (3.5) can be written in terms of $h_\varepsilon(s)$ as (refer to appendix A(b) for more details)

$$\Gamma_1(h_\varepsilon(s)) = \Delta_0 + \Delta_1 \left(\frac{e^{rt_c} s^{1/\varepsilon} (1-s)^{1/\varepsilon}}{s^{2/\varepsilon} + e^{2rt_c} (1-s)^{2/\varepsilon}} \right) + \Delta_2 \left(\frac{s^{2/\varepsilon} - e^{2rt_c} (1-s)^{2/\varepsilon}}{s^{2/\varepsilon} + e^{2rt_c} (1-s)^{2/\varepsilon}} \right) \quad (3.9)$$

and

$$\Gamma_2(h_\varepsilon(s)) = \delta_0 + \delta_1 \left(\frac{s^{2/\varepsilon} - e^{2rt_{c1}} (1-s)^{2/\varepsilon}}{s^{2/\varepsilon} + e^{2rt_{c1}} (1-s)^{2/\varepsilon}} \right) - \delta_1 \left(\frac{s^{2/\varepsilon} - e^{2rt_{c2}} (1-s)^{2/\varepsilon}}{s^{2/\varepsilon} + e^{2rt_{c2}} (1-s)^{2/\varepsilon}} \right), \quad (3.10)$$

and the new compactified autonomous system defined on the phase space $\mathbb{S}^n \times [0, 1]$ with the invariant subspaces $\{s = 0\}$ and $\{s = 1\}$ is given by ($i = 1, 2$)

$$\left. \begin{aligned} \dot{X} &= f(X, \Gamma_i(s)) \\ \dot{s} &= r\eta(s) = \frac{r}{h'_\varepsilon(s)} \end{aligned} \right\} \quad (3.11)$$

and

where

$$f(X, \Gamma_i(s)) = \begin{cases} f(X, \Gamma_i(h_\varepsilon(s))) & \text{for } s \in (0, 1), \\ f(X, \Gamma_i^-) & \text{for } s = 0, \\ f(X, \Gamma_i^+) & \text{for } s = 1 \end{cases}$$

and

$$\eta(s) = \frac{1}{h'_\varepsilon(s)} = \varepsilon s(1-s),$$

with the initial conditions $X(0) = X_0 \in \mathbb{S}^n$ and $s(0) = g_\varepsilon(0)$. The notation $'$ denotes d/ds . Further, the system (3.11) is guaranteed to be C^1 -smooth in the extended phase space by verifying the transformation conditions outlined in appendix A(c). With this autonomous compactified system, the analysis of R-tipping can be carried out by applying the tools of autonomous bifurcation theory, treating the rate parameter as an additional bifurcation parameter. A detailed illustration of this procedure is provided in the following sections using the savanna–forest system.

In essence, in the compactified formulation, equilibria of the past system at $s = 0$ acquire repelling trajectories that connect to equilibria of the future system at $s = 1$. These connections depend jointly on the rate and the magnitudes of the input. Put simply, the compactification converts a non-autonomous, time-dependent problem into a geometric problem of phase space connection, making it possible to use established dynamical system tools. Consequently, the R-tipping problem in the non-autonomous system (3.1) becomes equivalent to a problem of heteroclinic connections between equilibria in the past and future limit systems, with an

additional eigendirection along the transverse s -coordinate linking the invariant subspaces $\{s = 0\}$ and $\{s = 1\}$ in the compactified system (3.11). The heteroclinic orbit connecting the base equilibrium (the equilibrium from which tipping will be observed) in the past to a saddle equilibrium (edge state) in the future acts as the R-tipping threshold, separating the basins of attraction of the attractors in the compactified autonomous system. The corresponding rate defines the critical rate r_c . For more details on compactification and R-tipping, refer to [26,65].

4. Results

The savanna–forest ecological system (2.1) captures the full range of possible savanna ecosystem dynamics [54]. This section utilizes such a model to investigate the potential for rate-induced critical transitions in the compactified savanna–forest subsystems, each subjected to two distinct non-monotone external inputs. Each subsection begins with a bifurcation analysis, then explores BI and concludes with observations of R-tipping using two-dimensional tipping diagrams.

(a) Grass-driven dynamics in grassland–forest subsystem

A savanna ecosystem where interactions primarily occur between grassland and tropical forest trees, without significant presence of savanna trees, can be observed in the savanna–forest ecotones of the Amazon Basin in South America. In the absence of savanna trees, where $S = T = 0$, the dynamics in system (2.1) reduces to the grassland–forest sub-dynamics, which is given by a one-dimensional equation:

$$\dot{G} = \varphi(G)F - \alpha GF, \quad (4.1)$$

with $F = 1 - G$. Recall that the parameter α represents the recruitment rate of forest trees, whereas $\varphi(G)$ is an increasing sigmoidal function between φ_0 and φ_1 that shows how grass cover affects the mortality rate of forest trees owing to fire equation (2.3). In further discussions, the parameters α and φ_1 simultaneously vary over time to investigate R-tipping and are referred to as input parameters. The above-described autonomous system can have up to four steady states (equilibria). One of these is the grass-only equilibrium, $e_G = (1, 0)$, which signifies a state where the land can be completely occupied by grass cover without trees, whose stability depends on a wide range of parameters. Also, it is a common equilibrium across all subsystems of system (2.1). The other equilibria include stable high (e_H), stable low (e_L) grass cover and unstable (e_u) grass cover. They correspond to the intersection points of the sigmoidal function $\varphi(G)$ equation (2.3) and the line αG (see figure 3c). Depending on parameter settings, the system may exhibit bistable dynamics (the focus of this study) or monostable behaviour.

Figure 3a presents the two-parameter bifurcation or stability diagram for system (4.1), where the recruitment rate α and high fire-induced mortality rate φ_1 are varied. The diagram highlights two bistable regions—dark and light green—enclosed by saddle–node bifurcation curves SN_1 and SN_2 (black), which converge at a cusp point. In addition, a transcritical bifurcation curve (magenta) intersects SN_2 . Within the chosen parameter ranges, the bistable region I, characterized by the coexistence of high (e_H) and low (e_L) grass cover, is broader than region II, where the grass-only (e_G) and low (e_L) grass cover equilibria coexist. For a fixed mortality rate $\varphi_1 = 0.9$ (indicated by the horizontal black dotted line in figure 3a), one-parameter bifurcation (α versus G) has been plotted in figure 3b with a bistable region I shaded in light green. The blue and light green curves represent high (e_H) and low (e_L) grass cover equilibria, respectively. The solid dark green line denotes the stable grass-only equilibrium, which becomes and persists as unstable (dashed dark green line) after transcritical bifurcation (magenta-filled circle). The unstable equilibrium, e_u (grey dashed curve), intersects e_H and e_L at the fold points SN_1 and SN_2 (black filled circles), respectively. For the fixed parameters $\varphi_1 = 0.9$ (horizontal dotted line in figure 3a) and $\alpha = 1.2$ (horizontal dotted line in figure 3b), figure 3c,d is plotted with the same colour scheme representing equilibrium points. Figure 3d illustrates the phase portrait (G versus $\dot{G}$) with the vector field highlighting the stable equilibria e_L (green-filled circle) and e_H (blue-filled circle).

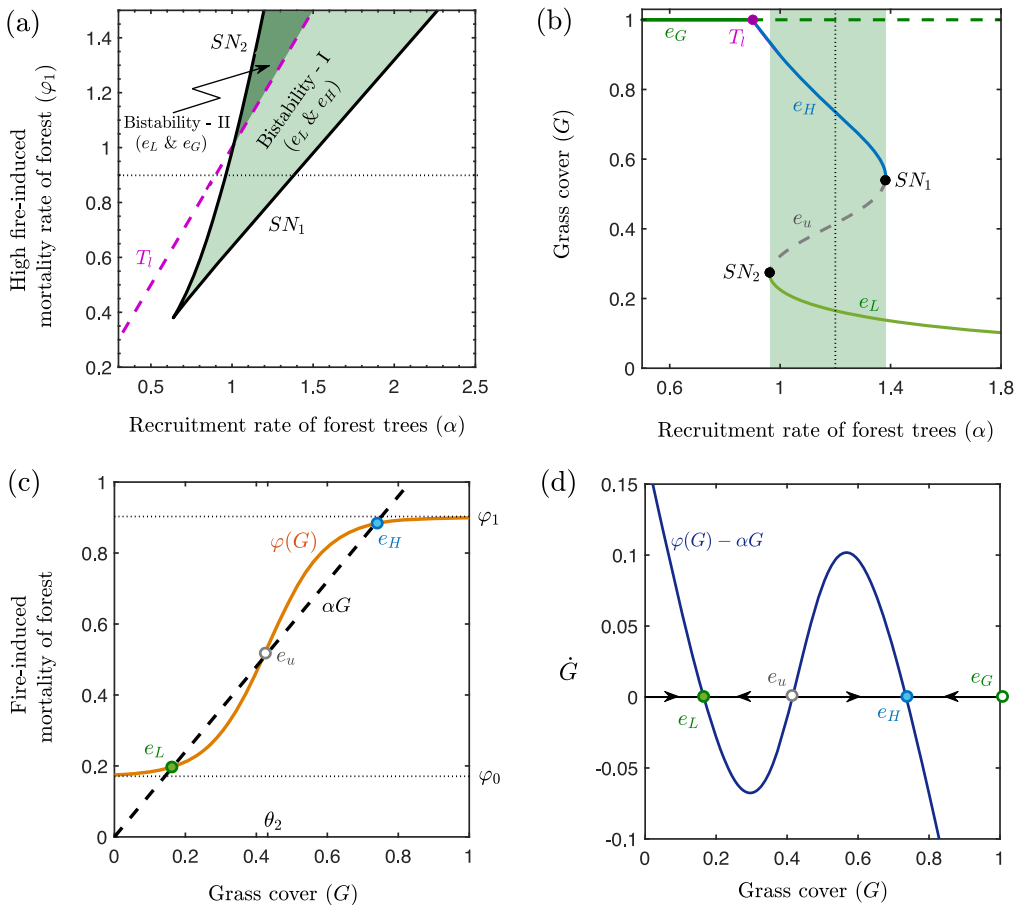


Figure 3. (a) Two-parameter bifurcation diagram illustrating the effects of the recruitment rate of forest trees (α) and the high mortality rate owing to fire (φ_1) in the grassland–forest subsystem (4.1). It highlights two bistable regions—Region I (light green) and Region II (dark green)—enclosed by saddle–node bifurcation curves SN_1 and SN_2 (black curves) and separated by a transcritical bifurcation curve T_l (magenta). (b) One-parameter bifurcation diagram plotted for a fixed $\varphi_1 = 0.9$ (indicated by a dotted line in (a)), showing the effect of α on the grass cover (G). The bistable Region I contains the stable equilibrium points e_L (light green curve) and e_H (blue curve). The unstable equilibrium points e_u (dashed grey curve) and e_G (dashed dark green line) are also shown, where e_G is stable before T_l (dark green line). The black and magenta-filled circles represent saddle–node and transcritical bifurcation points, respectively. (c) Equilibrium points in bistable Region I (excluding e_G) are determined by the intersection of the increasing sigmoidal function $\varphi(G)$ (orange curve), which varies between φ_0 and φ_1 and represents fire-induced mortality of forest trees, and the linear function αG (dashed black line). These curves correspond to fixed values $\varphi_1 = 0.9$ and $\alpha = 1.2$, denoted by black dotted lines in (a) and (b). (d) The vector field (G versus $\dot{G}$) with all equilibrium points. Other parameter values: $\varphi_0 = 0.1709$, $\theta_2 = 0.432$ and $s_2 = 0.082$.

(i) Rate-induced bifurcation in the compactified grassland–forest subsystem

We commence our analysis on BI in the bistable region I of the two-parameter bifurcation diagram (α versus φ_1) from figure 3a. In this subsystem, we observe that the low grass cover equilibrium e_L is basin unstable along a parameter path $\tilde{P}$, which is chosen as $\varphi_1 = k_0 + k_1\alpha + k_2\alpha^2$. Figure 4a illustrates that for the parameter path $\tilde{P}$ (dashed brown curve), which varies from p^- to p^+ , we can obtain a region of BI (grey-shaded) for the base equilibrium e_L . More clearly, we fix the attractor $e_L(p^-)$, where $p^- = (\alpha^-, \varphi_1^-)$, and vary the unstable equilibrium e_u along the path from p^- to p^+ and observe whether or not the attractor $e_L(p^-)$ enters into the basin of attraction of the high grass cover e_H . The collection of parameters p that results in such an entry defines the

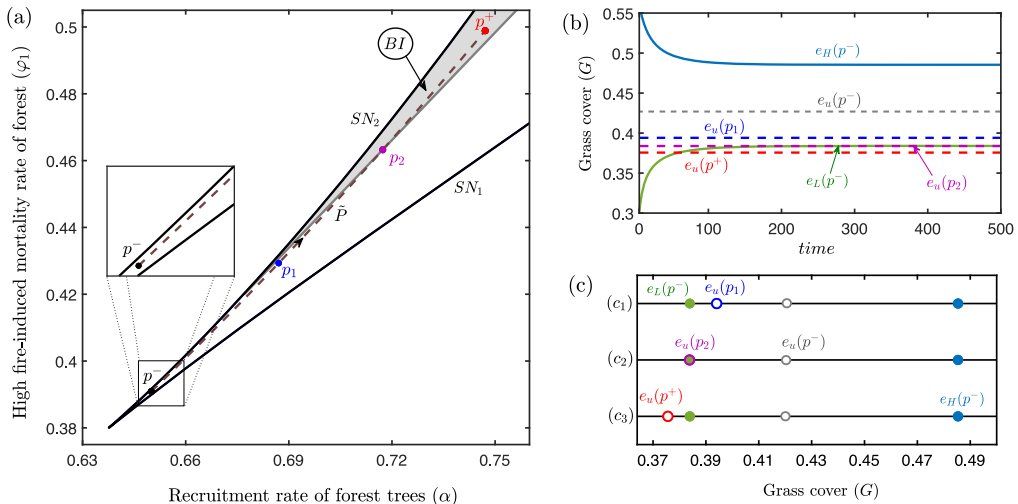


Figure 4. (a) The bistable region I of the two-parameter bifurcation diagram of [system \(4.1\)](#), highlighting the region of basin instability (BI) of the stable equilibrium e_L (grey-shaded). The dashed brown curve, varying between $p^- = (\alpha^-, \varphi_1^-) \approx (0.65, 0.3909)$ and $p^+ = (\alpha^+, \varphi_1^+) \approx (0.7471, 0.4989)$, represents a parameter path $\tilde{P}$: $\varphi_1(\alpha) = k_0 + k_1\alpha + k_2\alpha^2$, where $k_0 \approx 0.276$, $k_1 \approx -0.6367$ and $k_2 \approx 1.2514$. The zoomed-in picture illustrates that the parameter path does not intersect the saddle–node bifurcation curves. (b) Time series depicting trajectories converging to the stable equilibria e_L and e_H (solid green and blue curves) for a fixed input parameter p^- , together with an unstable equilibrium $e_u(p^-)$ (dashed grey line). Other dashed colour lines indicate unstable equilibria e_u for different input parameters. For $p = p_1 \approx (0.6871, 0.4293) \in \tilde{P}$, the unstable equilibrium e_u (blue dashed line) remains above $e_L(p^-)$, indicating no BI. However, for $p = p_2 \approx (0.7173, 0.4632)$ (dashed magenta line) and p^+ (dashed red line), the unstable equilibrium reaches and subsequently crosses $e_L(p^-)$, demonstrating the onset of BI. (c) Equilibrium points of [system \(4.1\)](#) on three lines (c_1) , (c_2) , (c_3) illustrate the transition of the stable equilibrium e_L from no instability to a BI state. Initially, e_L is not basin unstable (see the relative position of e_L and e_u on (c_1)), but as parameters progress through p_1 , p_2 and p^+ along the parameter path $\tilde{P}$ (following the colour scheme in (a)), BI emerges (see in (c_3) that, now e_u shifts in the left-hand side of e_L). The high grass cover stable equilibrium e_H (blue-filled circle) and the corresponding unstable equilibrium e_u (grey unfilled circle) are shown for the fixed parameter $p = p^-$.

region of BI associated with the fixed attractor $e_L(p^-)$. To demonstrate this, we plot time series and phase portraits of the frozen grassland–forest [subsystem \(4.1\)](#) by fixing all the equilibria corresponding to the parameter p^- and varying the unstable equilibrium corresponding to three distinct parameters p_1, p_2 and p_3 . Time series in [figure 4b](#) show three equilibria: e_H (solid blue), e_u (dashed grey) and e_L (solid green) for the fixed parameter p^- . As the parameters vary along $\tilde{P}$, the attractor $e_L(p^-)$ is initially not basin unstable with respect to $e_u(p_1)$, onsets for $e_u(p_2)$ and finally becomes basin unstable for $e_u(p^+)$. This progression is represented by dashed lines from blue, magenta and red in the time series, respectively. The corresponding phase portraits in [figure 4c](#) present the BI of e_L with no BI (see the relative position of e_L and $e_u(p^+)$ in (c_1)) and basin unstable in (c_2) and (c_3) . It is important to emphasize that although the parameter path is chosen to lie narrowly near the cusp point, it does not cross any fold bifurcation curves (see zoomed-in picture in [figure 4a](#)), ensuring the tipping will be due to the rate, not the magnitude, of the input parameters. Thus, tracing the parameter path $\tilde{P}$ over time ensures R-tipping when the rate exceeds some critical threshold.

Our analysis of rate-induced critical transitions in the grassland–forest subsystem focuses on two main fitness parameters of tropical forest trees: α and φ_1 , which vary over time. The parameters are considered where one depends on the other, as reflected in the quadratic form of the parameter path $\tilde{P}$. As the parameter path progresses from p^- to p^+ , the recruitment rate increases, leading to a corresponding rise in fire-induced mortality. This raises a natural question: can birth and fire-induced mortality rates increase simultaneously, or are they inherently

constrained by one another? Empirical evidence suggests that they can indeed rise together [34,72]. Trees with faster growth rates have a high risk of exposure to fire. In their rapid growth stages, they prioritize expansion over defense, which leads to smaller and thinner bark, making them more vulnerable to fire. In addition, increased growth rates can lead to denser forests, which might accumulate more fuel (litter, deadwood) and increase fire intensity and mortality. Since the fire-induced mortality rate of trees depends on grass cover, the rapid growth of trees could temporarily suppress grass (and thus fires). However, if fires occur, their intensity might increase owing to accumulated biomass, leading to higher mortality rates. In regions where the climate, such as rainfall, promotes tree growth, the same environment might also favour grasses, increasing fire intensity and, consequently, tree death rates. As a result, there is a high chance of transition from high forest cover (low grass cover) to low forest cover (high grass cover) if these input parameters vary above a critical rate and, hence, rate-induced critical transitions.

We begin our investigation by considering the compactified grassland–forest subsystem, as defined in system (3.11), where $G + F = 1$ and $S = T = 0$. First, we employ the external forcing term Γ_1 equation (3.4) to simultaneously vary both α and φ_1 , enabling us to examine R-tipping. The compactified grassland–forest subsystem, incorporating the modified input parameter $\alpha(s)$ equation (3.9), is formulated on $\mathbb{S}^1 \times [0, 1]$ as follows:

$$\begin{cases} \dot{G} = f(G, \alpha(s), \varphi_1(s)), \\ \dot{s} = r\eta(s), \end{cases} \quad (4.2)$$

where

$$f(G, \alpha(s), \varphi_1(s)) = \begin{cases} (\varphi(G, \varphi_1(s)) - \alpha(s)G)(1 - G) & \text{for } s \in (0, 1), \\ (\varphi(G, \varphi_1^-) - \alpha^- G)(1 - G) & \text{for } s = 0, \\ (\varphi(G, \varphi_1^+) - \alpha^+ G)(1 - G) & \text{for } s = 1 \end{cases} \quad (4.3)$$

and

$$\eta(s) = \varepsilon s(1 - s), \quad (4.4)$$

with $\varphi_1(s) = k_0 + k_1\alpha(s) + k_2(\alpha(s))^2$ and $\varepsilon \in (0, 2]$ denotes the compactification parameter. The main advantage of the compactified system is the presence of flow-invariant subspaces (topological), specifically:

$$S^- = \mathbb{S}^1 \times \{0\} \quad \text{and} \quad S^+ = \mathbb{S}^1 \times \{1\},$$

which contains the equilibrium points $e_j^\mp$, where $j = L, H, G, u$ from the autonomous past and future limit systems, respectively. When projected in the extended space of the compactified subsystem (4.2), these equilibrium points become

$$\tilde{e}_j^- = (e_j^-, 0) \quad \text{and} \quad \tilde{e}_j^+ = (e_j^+, 1),$$

for $j = L, H, G, u$. Each equilibrium point in this extended system acquires an additional eigendirection and, hence, the stable and unstable manifolds. These manifolds closely resemble the moving stable and unstable equilibria present in the original non-autonomous systems.

For the chosen compactification parameter value $\varepsilon = 0.75$, figure 5a presents a two-dimensional tipping diagram with the variation in the parameters of the external input $\alpha(s)$ following equation (3.9) (for more details about R-tipping using the other input Γ_2 from equation (3.10), refer to §S1.2 in the electronic supplementary material). Specifically, the diagram captures the influence of Δ_1 , which determines the peak height or magnitude of the input, and the rate parameter r , highlighting two primary regions: red and white, corresponding to R-tipping and tracking, respectively. Trajectories originating from and near the low grass cover equilibrium $\tilde{e}^-$, which is basin unstable in the frozen system, with the input pair (Δ_1, r) in the white region track the same

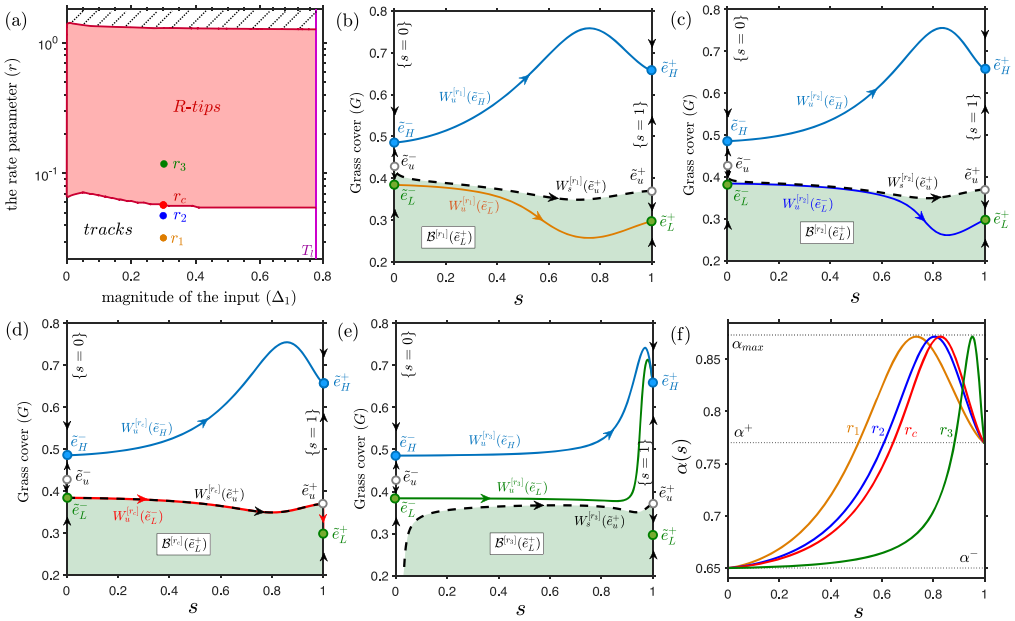


Figure 5. (a) Two-dimensional R-tipping diagram for the compactified grassland–forest subsystem (4.2), illustrating the regions of tipping (red) and tracking (white) as the rate parameter r and the parameter Δ_1 of the input equation (3.9) vary. The magenta line close to the right-hand boundary of these regions represents the magnitude $\Delta_1(T_I) = 0.774$ corresponding to the transcritical bifurcation T_I . The dotted area at the top indicates that for large values of the rate, the input pulse flattens into a horizontal line, resulting in discontinuities in the trajectories. For a fixed $\Delta_1 = 0.3$ and different r , subplots (b)–(e) illustrate the tracking and tipping behaviour in the compactified phase space (s versus G), where the invariant subspaces $\{s = 0\}$ and $\{s = 1\}$ contain the equilibria $\tilde{e}_j^\pm$ for $j = L, H, u, G$ from the past and future limit systems. Within the extended phase spaces, the solid blue curve at the top represents the invariant unstable manifold $W_u^{[r]}(\tilde{e}_H^-)$, which links the past high grass cover equilibrium $\tilde{e}_H^-$ to the future equilibrium $\tilde{e}_H^+$. The green-shaded area represents the basin of attraction $\mathcal{B}^{[r]}(\tilde{e}_L^+)$ for the future stable low grass cover equilibrium $\tilde{e}_L^+$, with the basin boundary $W_s^{[r]}(\tilde{e}_u^+)$ (dashed black curve) acting as the R-tipping threshold. As the rate varies, this $\mathcal{B}^{[r]}(\tilde{e}_L^+)$ contracts. (b) For $r_1 = 0.032$, the manifold $W_u^{[r_1]}(\tilde{e}_L^-)$ (orange curve) that approximates the moving stable low grass cover equilibrium $e_L(rt)$ in the original non-autonomous system and tracks the same and accumulates in the future equilibrium $\tilde{e}_L^+$. (c) For $r_2 = 0.05$, the R-tipping threshold $W_s^{[r_2]}(\tilde{e}_u^+)$, and the manifold $W_u^{[r_2]}(\tilde{e}_L^-)$ (blue curve) come closer. (d) At the critical rate $r_c \approx 0.0572$, these manifolds merge (red curve), forming the heteroclinic trajectory $\tilde{e}_L^-$ to $\tilde{e}_u^+$, marking the onset of R-tipping. (e) When the rate increases further to $r_3 = 0.12$, the manifolds separate and the manifold from $W_u^{[r_3]}(\tilde{e}_L^-)$ accumulates at the future high grass cover equilibrium $\tilde{e}_H^+$, signifying the occurrence of R-tipping. (f) The variation of the input parameter α is shown concerning the added dimension s for the different rates used, following the same colour scheme as in (a). The input starts at $\alpha^- = 0.65$, reaches a peak at $\alpha_{\max} = 0.8714$, and eventually settles at $\alpha^+ = 0.77$ (dotted lines). The compactification parameter is set to $\varepsilon = 0.75$ for this analysis. Additional parameter values for the input include $\Delta_0 = 0.71$, $\Delta_2 = 0.3$ and $t_c = 30$.

in future invariant subspace $\tilde{e}^+$, whereas for the pair (Δ_1, r) in the red region, the trajectories tips to another attractor in the future invariant subspace. The peak of an input reaches a transcritical bifurcation point T_I (marked in magenta) at the magnitude $\Delta_1(T_I)$. The dotted region above the R-tipping zone corresponds to the stiffness of the input when the rate parameter exceeds a large value. More specifically, for very large values of the rate, the input flattens into a horizontal line, resulting in a discontinuity in the trajectories before they reach the equilibrium in the future invariant subspace.

More evidently, the phase portraits in figure 5b–e, the past invariant subspace $\{s = 0\}$ contains the saddle low grass equilibrium $\tilde{e}_L^-$ (green filled circle), which is stable within the subspace and unstable in the transverse s -direction. The invariant subspace $\{s = 1\}$ contains the stable equilibria $\tilde{e}_L^+$ (green filled circle) and $\tilde{e}_H^+$ (blue filled circle), together with the saddle $\tilde{e}_u^+$ (unfilled circle),

known as the R-tipping edge state. The R-tipping threshold (dashed black curve) is the stable invariant manifold $W_s^{[r]}(\tilde{e}_u^+)$ whose shape depends on the rate r and magnitude Δ_1 . This threshold separates the phase space into two basins of attraction, with the basin of attraction $\mathcal{B}^{[r]}(\tilde{e}_L^+)$ shaded in green. For a fixed magnitude Δ_1 and a lower rate $r = r_1$ in figure 5b, the unstable manifold $W_u^{[r_1]}(\tilde{e}_L^-)$ (orange) closely approximates the moving stable equilibrium $e_L(rt)$, while the R-tipping threshold $W_s^{[r_1]}(\tilde{e}_u^+)$ closely approximates the moving unstable equilibrium $e_u(rt)$. The trajectories started from and near $\tilde{e}_L^-$ accumulate on $\tilde{e}_L^+$. On further increasing the rate to r_2 in figure 5c, both $W_u^{[r_2]}(\tilde{e}_L^-)$ (blue) and $W_s^{[r_2]}(\tilde{e}_u^+)$ get close to each other, and the size of the basin of attraction $\mathcal{B}^{[r_2]}(\tilde{e}_L^+)$ starts shrinking. At a critical rate $r \approx r_c$, there is a heteroclinic connection (see figure 5d) from $\tilde{e}_L^-$ to $\tilde{e}_u^+$, given by the intersection of the invariant manifolds $W_u^{[r_c]}(\tilde{e}_L^-)$ (red) and $W_s^{[r_c]}(\tilde{e}_u^+)$, allowing some trajectories in the neighbourhood of $\tilde{e}_L^-$ to track the same as well as tip to an alternative attractor in the future invariant subspace. Figure 5e summarizes the occurrence of R-tipping to an alternative high grass $\tilde{e}_H^+$ equilibrium in the future invariant space, failing to track the moving stable equilibrium $e_L(rt)$ in the non-autonomous system. For the rate value $r = r_3$, which is inside the tipping region in figure 5a, the unstable manifold $W_u^{[r_3]}(\tilde{e}_L^-)$ (dark green) accumulates on $\tilde{e}_H^+$, meaning the trajectories in the selected open neighbourhood around $\tilde{e}_L^-$, all evolve and converge to the other stable equilibrium $\tilde{e}_H^+$. In all the figures, the blue curve on the top represents an unstable manifold $W_u^{[r]}(\tilde{e}_H^-)$ from $\tilde{e}_H^-$ which connects to $\tilde{e}_H^+$. Finally, figure 5f shows the distinct inputs α concerning the extended dimension s for different rate values following the same colour code from other figures. Fig. S1 in §S1.1 of the electronic supplementary material demonstrates R-tipping of the compactified grassland–forest subsystem (4.2) on the autonomous cusp surface using the input Γ_1 equation (3.9).

(b) Dominance of woodland over grassland in the absence of forest trees

Another sub-dynamic inferred from the global savanna–forest model is the grassland–woodland (also simply referred to as the woodland) subsystem. Adult savanna trees differ from tropical forest trees in morphological and physiological features; the bark of savanna trees is thicker than that of forest trees, making it fire-resistant to frequent fires [47]. Savanna saplings have a high root-to-shoot ratio with ample underground storage, making them more susceptible to top-kill during fires compared to forest tree saplings. The grassland–woodland subsystem describes the competition between grasses and adult savanna trees without forest trees ($G + S + T = 1, F = 0$) [54]. It also includes savanna saplings as one of the state variables, making the system a stage-structured model and highlighting its essential contribution to the dominance of savanna trees over grass. These types of ecosystems can be found in East African savannas, where the annual rainfall ranges between 51 and 127 cm, which is insufficient to support denser tropical forests, and the soil fertility is also poor.

To keep the investigation simple, we restrict the model by neglecting the influence of savanna saplings (S) and adult savanna trees (T) on fire spread, setting $\gamma = 0$ in system (2.1) for the remainder of the analysis. When the forest tree functional type F vanishes, the system (2.1) reduces to a two-dimensional subsystem that interplays between grass, savanna saplings and adult savanna trees. The autonomous grassland–woodland subsystem defined on $\mathbb{S}^2$ is as follows:

$$\text{and} \quad \left. \begin{aligned} \dot{G} &= \mu S + \nu T - \beta GT \\ \dot{T} &= \omega(G)S - \nu T, \end{aligned} \right\} \quad (4.5)$$

with $S = 1 - G - T$.

To be recalled, the parameters μ and ν refer to the mortality rates of the savanna saplings and adult savanna trees, respectively. Savanna saplings are born at the rate of β when grasses and adult savanna trees interact. The function $\omega(G)$ (see equation (2.2)) is the decreasing sigmoidal

sapling to adult tree maturation rate from ω_0 to ω_1 . Depending on parameter values, the system will have at most four equilibrium points, such as f_L and f_H , denoting low and high adult savanna tree cover, respectively, and separated by an unstable one f_u in the bistable zone. As mentioned earlier, the grass-only equilibrium $e_G = (1, 0, 0)$ is also present in this subsystem.

Fig. S3(a) in §S2.1 of the electronic supplementary material illustrates the two-parameter bifurcation diagram (β versus ω_1) of system (4.5). Similar to the one-dimensional subsystem (4.1), we have two bistable regions, A (light green) and B (dark green), enclosed by two saddle-node bifurcations (black) curves SN_3 and SN_4 . The brown curve, which separates both bistable regions, is a transcritical bifurcation curve TL_1 . Region A contains two stable (f_L and f_H) and one saddle (f_u) equilibrium point, the region of interest. For a fixed value of ω_1 , one-parameter bifurcation (β versus G) can be seen in the sub-figure (b), with the bistable region A shaded green. The blue and green curves represent high (f_H) and low (f_L) savanna forest cover, respectively. The dashed grey curve between them is a saddle equilibrium (f_u). The dark green line on the top is the grass-only equilibrium (e_G).

(i) Rate-induced tipping in the woodland subsystem

Once again, we examine for BI of an attractor in the autonomous grassland–woodland frozen subsystem (4.5) using the input parameters β and ω_1 to guarantee R-tipping in the corresponding non-autonomous system where these parameters are varied over time. In this case, the low adult savanna tree cover equilibrium f_L is basin unstable for the parameter path $\tilde{Q}$. This can be seen in detail in figure 6. For the parameter path $\tilde{Q}$ varying from q^- to q^+ , figure 6a showcases the region of BI (grey shaded) of the base stable equilibrium f_L . The 2-simplex $\mathbb{S}^2$ phase portraits in figure 6b–d presents the BI of f_L . First, we fix the equilibrium points f_L, f_H (dark filled circles), and f_u, e_G (black unfilled circles) for the parameter $q^- = (\beta^-, \omega_1^-)$ and change the equilibrium points f_L, f_H, f_u corresponding, respectively, to parameters q_1, q_2 and q^+ on the parameter path $\tilde{Q}$. The blue shaded region highlights the presence of basin-unstable stable equilibrium $f_L(q^-)$. For the parameter, $q = q_1$, the stable equilibrium f_L shows no BI without crossing the stable manifold $w^s(f_u)$ of the saddle equilibrium $f_u(q_1)$ (orange) in figure 6b, whereas $f_L(q^-)$ is basin unstable (see figure 6c,d) by intersecting and crossing the stable manifolds of $f_u(q_2)$ (magenta) and $f_u(q^-)$ (red).

Adult savanna trees and their saplings play a crucial role in maintaining the balance of the savanna–forest ecosystem by reducing grass fuel loads and influencing fire dynamics [31,46]. Savanna saplings are fire-adapted and can resprout readily after fires, ensuring the persistence of savanna trees despite frequent fires [73]. Some species have higher survival rates in areas prone to fire by producing thick bark early (e.g. African baobab, acacia and combretum species). With this significance, here we study rate-induced critical transitions between low and high savanna forest cover. To achieve this, we vary two input parameters through the parameter path $\tilde{Q}$ from the autonomous bifurcation diagram (see figure 6). Specifically, we increase the recruitment rate of savanna trees β and decrease the lower maturation rate owing to fire ω_1 . Although such a variation is specific and not particularly common, it is still possible to observe such a change [34,72]. A lower ω_1 means that fire more strongly suppresses sapling to adult tree maturation at high grass, reducing the number of saplings that transition into mature savanna trees [41]. However, if adult savanna tree recruitment rates increase significantly, more saplings enter the system to offset fire-induced losses [46]. With the above argument, we begin our analysis of R-tipping in the following compactified autonomous system by incorporating the recruitment rate as $\beta(s)$, following equation (3.9), and defined on $\mathbb{S}^2 \times [0, 1]$. The compactified grassland–woodland subsystem is given by

$$\left. \begin{aligned} \dot{G} &= f(G, T, \beta(s), \omega_1(s)), \\ \dot{T} &= g(G, T, \beta(s), \omega_1(s)) \\ \dot{s} &= r\eta(s), \end{aligned} \right\} \quad (4.6)$$

and

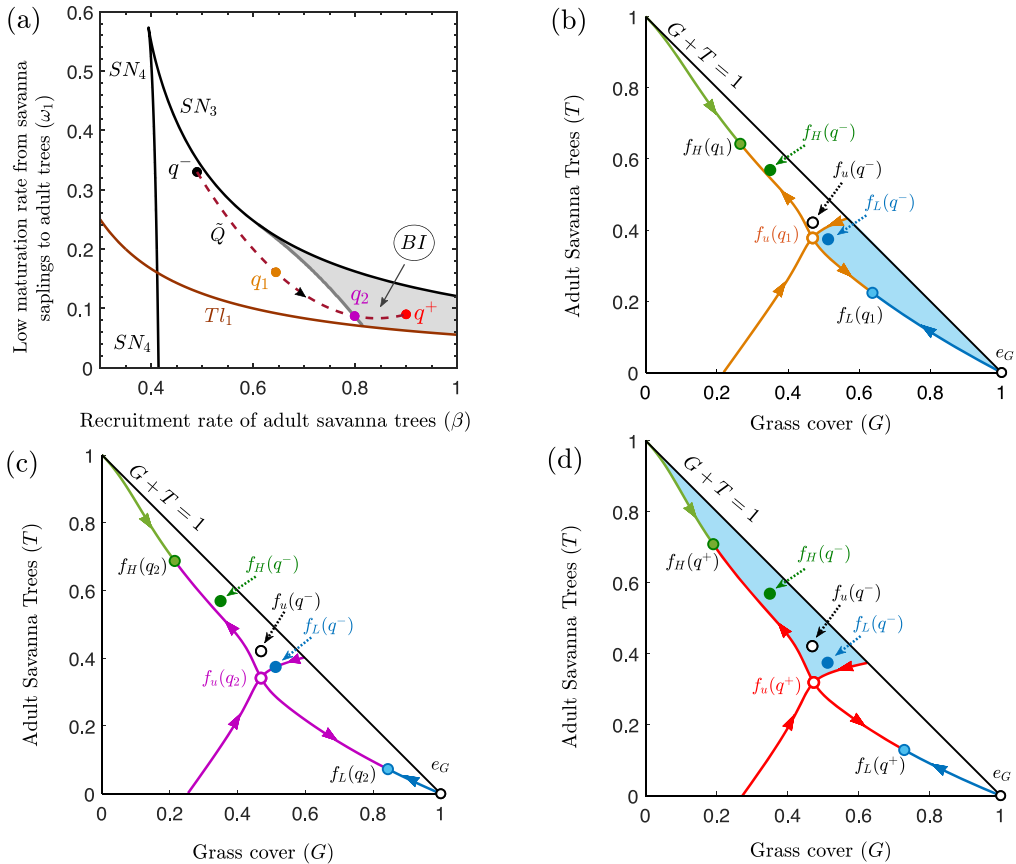


Figure 6. (a) The grey-shaded region represents the basin BI of the low savanna forest equilibrium, f_L , within the bistable region A of the grassland–woodland subsystem (4.5) in the β – ω_1 bifurcation diagram. The BI region is determined for the fixed parameter pair $q^- = (\beta^-, \omega_1^-) = (0.49, 0.33)$. The subsequent phase portraits illustrate the BI of f_L for three different parameter values, q_1 (orange), q_2 (magenta) and q^+ (red), along a nonlinear parameter path $\tilde{Q}$. This path, shown as a dashed brown curve, follows the equation $\omega_1 = l_0 + l_1\beta + l_2\beta^2$ where $l_0 \approx 1.494$, $l_1 \approx -3.349$ and $l_2 \approx 1.9882$. The 2-simplex phase portraits display two sets of equilibria: one corresponding to the fixed parameter q^- (depicted by dark-filled colours and black unfilled circles) and another set for q_1 , q_2 and q^+ (represented by light-filled colours and unfilled circles, following the same colour scheme as in (a)). The blue-shaded area indicates the presence of the equilibrium $f_L(q^-)$. The curves with arrows denote stable and unstable manifolds associated with saddle equilibrium points for the three parameter values, following the same colour scheme as in (a); crossing the stable manifold signifies BI. (a) At $q_1 \approx (0.64471, 0.1609)$, the stable equilibrium $f_L(q^-)$ is not basin unstable. (b)–(c) For parameter values $q_2 \approx (0.7991, 0.08706)$ and $q^+ = (0.9, 0.09)$, the equilibrium $f_L(q^-)$ first begins to exhibit BI and ultimately becomes basin unstable.

where

$$f(G, T, \beta(s), \omega_1(s)) = \begin{cases} \mu(1 - G - T) + \nu T - \beta(s)GT & \text{for } s \in (0, 1), \\ \mu(1 - G - T) + \nu T - \beta^- GT & \text{for } s = 0, \\ \mu(1 - G - T) + \nu T - \beta^+ GT & \text{for } s = 1, \end{cases} \quad (4.7)$$

$$g(G, T, \beta(s), \omega_1(s)) = \begin{cases} \omega(G, \omega_1(s))(1 - G - T) - \nu T & \text{for } s \in (0, 1), \\ \omega(G, \omega_1^-)(1 - G - T) - \nu T & \text{for } s = 0, \\ \omega(G, \omega_1^+)(1 - G - T) - \nu T & \text{for } s = 1 \end{cases} \quad (4.8)$$

and

$$\eta(s) = \varepsilon s(1 - s), \quad (4.9)$$

with $\omega_1(s) = l_0 + l_1\beta(s) + l_2(\beta(s))^2$ and $\varepsilon \in (0, 2]$ denotes the compactification parameter.

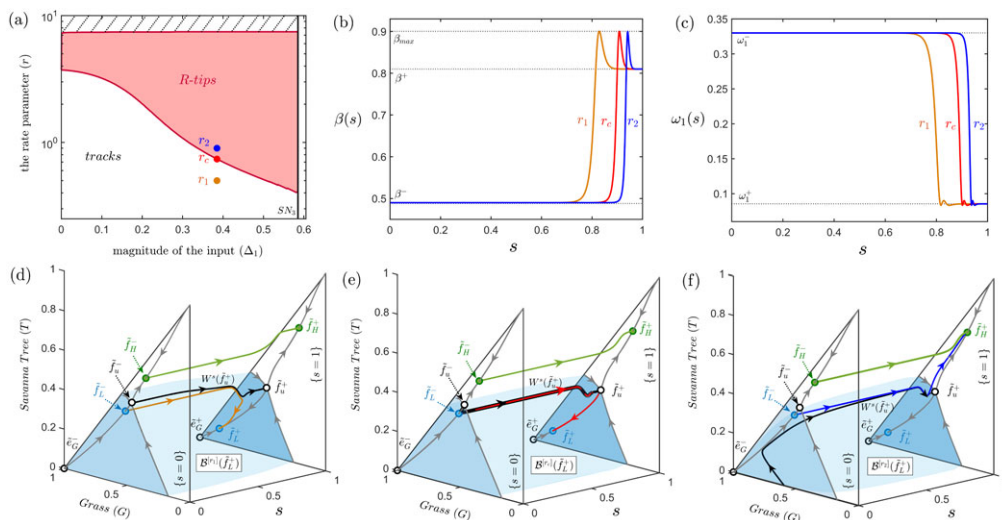


Figure 7. (a) Two-dimensional tipping diagram (Δ_1 versus r) for the compactified grassland–woodland subsystem (4.5) with the input equation (3.9). The red and white regions correspond to the R-tipping and tracking zones, respectively. These regions terminate at the saddle–node bifurcation input value $\Delta_1(SN_3) = 0.5853$. The remaining figures are plotted for a fixed value of $\Delta_1 = 0.385$ and three different rates (r_1, r_c and r_2). (b,c) External inputs— $\beta(s)$ and $\omega_1(s)$ —are shown for three different rate values, following the same colour scheme as in (a). (d)–(f) The extended phase portraits of system (4.6) illustrate tipping and tracking trajectories (manifolds) from the subspace $\{s = 0\}$ to the subspace $\{s = 1\}$. The dark blue shaded region in the invariant subspaces represents the basin of attraction of the low savanna forest equilibrium f_L in the original 2-simplex past and future limit system (4.5). The transparent blue region $\mathcal{B}^{nl}(\tilde{f}_L^+)$ between the invariant subspaces denotes the basin of attraction for the future equilibrium $\tilde{f}_L^+$ in the extended phase space, which depends on r . (d) At $r_1 = 0.5$, the unstable manifold $W_u^{nl}(\tilde{f}_L^-)$ (orange) from $\tilde{f}_L^-$ accumulates at the future equilibrium $\tilde{f}_L^+$, representing the tracking of the moving equilibrium $f_L(rt)$ in the original non-autonomous grassland–woodland subsystem. (e) At the critical rate $r_c \approx 0.739029$, the unstable manifold $W_u^{nl}(\tilde{f}_L^-)$ (red) nearly aligns with the R-tipping threshold $W_s^{nl}(\tilde{f}_L^+)$, forming a heteroclinic connection $\tilde{f}_L^-$ to $\tilde{f}_L^+$. (f) For $r_2 = 0.9$, the unstable manifold $W_u^{nl}(\tilde{f}_L^-)$ (blue) leads to tipping towards the future high savanna forest equilibrium $\tilde{f}_H^+$, indicating that $f_L(rt)$ tips to $f_H(rt)$ when $r = r_2$. For all three rate values, the basins of attraction $\mathcal{B}^{nl}(\tilde{f}_L^+)$ shrink, and the R-tipping threshold $W_s^{nl}(\tilde{f}_L^+)$ (black curves) lies on these regions. Other parameter values are: $\varepsilon = 0.1$, $\Delta_0 = 0.65$, $\Delta_2 = 0.16$ and $t_c = 30$.

Similar to the one-dimensional case, the extended phase space of system (4.6) contains the invariant subspaces

$$R^- = \mathbb{S}^2 \times \{0\} \quad \text{and} \quad R^+ = \mathbb{S}^2 \times \{1\},$$

with the equilibrium points from the autonomous past and future limit systems in the form of

$$\tilde{f}_j^- = (f_j^-, 0) \quad \text{and} \quad \tilde{f}_j^+ = (f_j^+, 1),$$

for $j = L, H, u$, respectively.

Figure 7a explains the rate-induced critical transition with a tipping diagram by varying the magnitude Δ_1 and the rate r of an input equation (3.9). The red and white regions correspond to R-tipping and tracking, respectively. After a particular magnitude value, the regions converge to a magnitude corresponding to the saddle–node bifurcation SN_3 (black line). For a fixed magnitude Δ_1 and three distinct rate values, we present three compactified phase spaces in figure 7d–f with unstable manifolds that approximate the moving stable equilibrium in the corresponding non-autonomous system. Since f_L is basin-unstable in the autonomous frozen system (4.5), we believe that we shall observe R-tipping from f_L . More clearly, in figure 7d, for the smaller rate value $r = r_1$, we observe that the unstable manifold $W_u^{nl}(\tilde{f}_L^-)$ (orange curve) originating from $\tilde{f}_L^-$ in the

past invariant subspace $\{s = 0\}$ accumulates at $\tilde{f}_L^+$ in the future space $\{s = 1\}$, whereas the R-tipping threshold $W_s^{[r_1]}(\tilde{f}_u^-)$ (black curve) from $\tilde{f}_u^-$ accumulates at the edge state $\tilde{f}_u^+$, illustrating the tracking of the moving stable equilibrium $f_L(rt)$. The transparent blue-shaded region in between represents the basin of attraction $\mathcal{B}^{[r_1]}$ of the future equilibrium $\tilde{f}_L^+$ bounded by the R-tipping threshold $W_s^{[r_1]}(\tilde{f}_u^+)$. As the rate value increases, this region shrinks, as further figures show. At a critical rate value $r \approx r_c$ (red curve), both the R-tipping threshold $W_s^{[r_c]}(\tilde{f}_u^+)$ and unstable manifold $W_u^{[r_c]}(\tilde{f}_L^-)$ merge together and form a heteroclinic connection between $\tilde{f}_L^-$ and $\tilde{f}_u^+$ (see figure 7e). And in further increasing $r = r_3$ (from the tipping region), we see in figure 7f, the trajectories originated on and near $\tilde{f}_L^-$ accumulate at $\tilde{f}_H^+$ (blue curve), indicating R-tipping from low forest cover to high forest cover in the non-autonomous grassland–woodland subsystem. In all the cases, high forest cover equilibrium tracks the same connecting past to future invariant subspaces by an unstable manifold $W_u^{[r]}(\tilde{f}_H^-)$ (green curve). Figure 7b,d corresponds to the increasing recruitment rate $\beta(s)$ and decreasing low maturation rate $\omega_1(s)$ for three distinct rate values following the same colour code as in figure 7a. The curves are translated to the right-hand side owing to the value of the peak fixed parameter t_c . For more details, see the two-dimensional projected version of figure 7 (refer to fig. S4 in §S2.2 and fig. S5 for the other input Γ_2 in §S2.3 of the electronic supplementary material).

(c) Full system dynamics: from steady-state coexistence to oscillations

This section examines R-tipping from a coexistence equilibrium, where all species coexist, to an oscillatory state in the full savanna–forest compactified system (fig. S6 in §S3.1 of the electronic supplementary material presents the bifurcation diagrams along with tetrahedral phase spaces of the full system).

We begin with a bifurcation analysis of the full savanna–forest autonomous system (2.1), subject to the constraint $G + S + T + F = 1$, with the variation of recruitment rates of forest and adult savanna trees. Fig. S6(a) in the electronic supplementary material presents a one-parameter bifurcation diagram (α versus G) that reveals multiple dynamical behaviours. Solid curves denote stable equilibria in this diagram, while dotted and dashed curves indicate saddle and unstable equilibria, respectively. As in the one-dimensional subsystem subsystem (4.1), saddle–node (SN_1 and SN_2) and a transcritical (TI) bifurcations are observed with an additional transcritical bifurcation (TI_2). Unstable limit cycles (orange dotted curves) emerge from two subcritical Hopf bifurcations (H_{sub}^1 and H_{sub}^2) and eventually merge and disappear at fold bifurcations of limit cycles (Flc_1 and Flc_2) marked by orange filled circles. Later, these fold bifurcations give rise to stable limit cycles (represented by the dark green curve). For a set of parameter values, we can observe the coexistence of a steady state and a stable limit cycle, separated by an unstable limit cycle (green shaded), which is the region of our interest. In this region, another stable equilibrium point exists owing to $S = T = 0$, which is the low grass cover equilibrium e_L from one-dimensional equation (4.1). Fig. S6(b) in electronic supplementary material provides a detailed view of the bifurcation diagram with an added dimension of adult savanna trees (T) on the z-axis.⁵ Figs. S6(c) and S6(d) in electronic supplementary material display the tetrahedral phase spaces (in different angles) with the dynamics of coexisting limit cycles and equilibrium points from the green shaded region of the bifurcation diagram.

(i) Rate-induced tipping in the savanna–forest (full) ecosystem

In this section, we analyse BI of the coexistence equilibrium and proceed further with the observation of R-tipping in the savanna–forest system.

As mentioned earlier, our region of study features the coexistence of a stable limit cycle Φ and a stable equilibrium point g_c , which are separated by an unstable limit cycle Θ (refer to fig. S7

⁵For more detailed bifurcation analysis of the savanna–forest system (2.1), refer to [54].

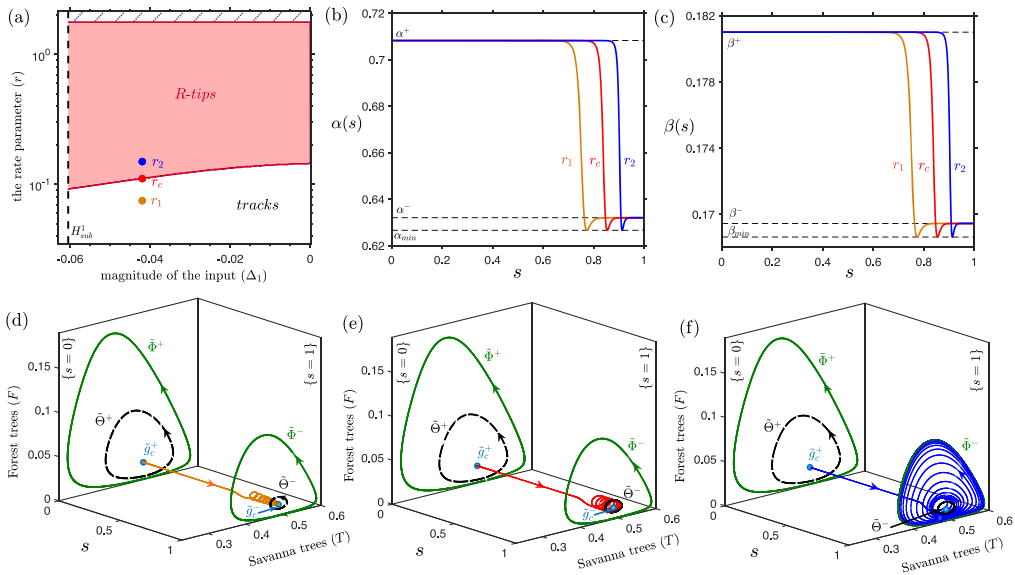


Figure 8. (a) The tipping diagram of the compactified savanna–forest (full) system (eqs. (S1)–(S5) in §S3.3 of electronic supplementary material), illustrating the regions of tipping (red) and tracking (white) concerning the external input equation (3.9). These regions are constrained on the left-hand side by a subcritical Hopf bifurcation $\Delta_1(H_{sub}^1) = -0.0605$ (black dashed line). The subsequent figures display phase portraits for a fixed magnitude $\Delta_1 = -0.042$ and three different rate values: r_1, r_c and r_2 . (b,c) The variation of external inputs $\alpha(s)$ and $\beta(s)$ is depicted for the three rate values, following the same colour scheme as in (a). (d) At the rate $r_1 = 0.075$, the phase portrait shows a trajectory originating from $\tilde{g}_c^-$ in the past invariant subspace $\{s = 0\}$, which continues to track the same state (orange curve) in the future invariant subspace $\{s = 1\}$. (e) For the critical rate $r_c \approx 0.11098$, the trajectory terminates at the unstable limit cycle $\tilde{\Theta}^+$ (red curve) in the future invariant subspace $\{s = 1\}$, known as edge state. (f) At a higher rate $r_2 = 0.15$, the trajectory tips to an alternative stable limit cycle $\tilde{\Theta}^+$ in the future invariant subspace $\{s = 1\}$ (blue curve). Other parameters: $\varepsilon = 0.075$, $\Delta_0 = 0.6700995$, $\Delta_2 = -0.0380995$ and $t_c = 200$.

of §S3.2 in the electronic supplementary material, which demonstrates the findings of BI of the equilibrium g_c).

When the recruitment rates of both adult savanna trees (β) and forest trees (α) decline, one can expect the possibility for the system to transition from a coexistence equilibrium (where all species are in steady-state balance) to an oscillation [54]. Intuitively, this can be interpreted as when tree recruitment rates (both) are high, a stable balance between G, S, T and F is well maintained [46]. As they decrease, both savanna and forest tree cover decline, leading to more available grass, which fuels fires [41]. Increased fire intensity further suppresses tree regeneration, enhancing oscillations in all functional types. With this understanding, we begin our analysis of R-tipping from steady state to oscillations by varying both the input parameters on the parameter path $\mathcal{U}$ from the bifurcation diagram (refer to fig. S7(a) in the electronic supplementary material). We utilize the external input Γ_1 equation (3.4) for the variation of parameters, but in descending structure.

Figure 8 presents the R-tipping analysis with the tipping diagram and phase portraits in the compactified autonomous system (see eqns. (S1)–(S5) in §S3.3 of the electronic supplementary material) defined on $\mathbb{S}^3 \times [0, 1]$. Figure 8a refers to the tipping diagram with the variation of parameters in the input equation (3.9): Δ_1 and r . Since the input parameters are declining, the magnitudes on the x -axis are negative. The red and white regions indicate the pairs (Δ_1, r) that result in R-tipping and tracking the attractors, respectively. The regions are bounded to the left-hand side by a magnitude value corresponding to the subcritical Hopf bifurcation. We consider three rates and present three-dimensional phase portraits to further demonstrate this, illustrating

tracking and tipping from the past invariant subspace. Though the original phase space is a 3-simplex (tetrahedron), with the extended variable, the system turns out to be four-dimensional. Figure 8d–f portrays the phase space with the past $\{s = 0\}$ and future $\{s = 1\}$ invariant subspaces containing the limit cycles $\tilde{\Phi}^\mp$, $\tilde{\Theta}^\mp$ and equilibrium points $\tilde{g}_c^\mp$. For the smaller rate value $r = r_1$, the unstable manifold $\tilde{W}^u(\tilde{g}_c^-)$ from $\tilde{g}_c^-$ accumulates at $\tilde{g}_c^+$ (orange curve), illustrating the tracking of the moving equilibrium $g_c(rt)$ in the corresponding non-autonomous system. At the critical rate $r \approx r_c$, the unstable manifold accumulates at a future unstable limit cycle $\tilde{\Theta}^+$ (red curve). In contrast, for high rate value $r = r_2$, the unstable manifold deviates from the future unstable limit cycle and accumulates at the future stable limit cycle $\tilde{\Phi}^+$ (blue curve), indicating that the trajectory fails to track the moving equilibrium $g_c(rt)$ and tips to the moving stable limit cycle $\Phi(rt)$ in the non-autonomous system. Figure 8b,c indicates the declining input parameters $\alpha(s)$ and $\beta(s)$, respectively, corresponding to the rates r_1 , r_c and r_2 following the same colour code. For the tipping diagram corresponding to the other input Γ_2 equation (3.10), refer to §S3.4 in the electronic supplementary material.

5. Conclusion and discussion

Savannas, which are grass-dominated ecosystems with sparse tree cover, are distinct and diverse biomes increasingly threatened by climate change and human interventions. The recent Global Tipping Points Reports 2023 [74] and 2025 [4] identify savannas as one of the potential biosphere tipping elements with medium confidence at local-to-landscape scales. In many regions, savannas and tropical forests act as natural alternative stable states [40]. Being the zone of transition between two very different biomes, savanna–forest transitions are expected to be highly sensitive to change, whether driven by global drivers such as rainfall and atmospheric carbon dioxide, or local drivers such as fire and herbivory regimes [34]. While the Global Tipping Points Reports assessments emphasize that tipping in savannas is likely to unfold over decades, resulting in relatively low confidence in abrupt, large-scale, rate-driven change, this does not preclude the possibility that rapid changes in ecological drivers may trigger transitions through rate-dependent mechanisms, which are often overlooked by assessments focused solely on static thresholds (bifurcations) [40].

Here, we demonstrate that savanna–forest ecosystems can undergo abrupt transitions driven by the rate at which key ecological processes are altered. There is ample empirical evidence that savanna–forest ecosystems are undergoing rapid changes globally, with woody plants increasingly invading grasslands in many regions [31]. Much of the literature on savanna–forest transitions focuses on crossing thresholds associated with bifurcations by varying parameters, including mean annual precipitation [35] and cluster density of forest trees [48]. Our results complement this perspective by showing that it is not just the crossing of critical levels leading to ecosystem collapse; the rate at which these changes occur also poses a serious threat to the stability of savanna–forest ecosystems [11,75,76]. To explore this, we employed a non-autonomous mathematical model that captures the coupled dynamics of grass, savanna saplings, adult savanna trees and forest trees, and investigated R-tipping by varying fitness parameters, such as recruitment, maturation and fire-induced mortality, in the form of bi-asymptotically constant inputs with sudden impulses, over time. These time-dependent impulses capture general features of short-lived disturbances, such as their rapid onset and subsequent recovery. In addition, we constrained the model so that the sum of all functional types remains constant, reducing the analysis from a regular Euclidean phase space into a simplex phase space.

The savanna–forest model provides a broadly applicable framework that captures the dynamics of all four functional types across different regions of the world. In some regions, primarily grasses and forest trees interplay, whereas in other regions, grasses coexist with savanna saplings and adult savanna trees. To address this ecological diversity in a unified modelling framework, we divided the global savanna–forest system into three subsystems with grass as the primary functional type [54]. Across these subsystems, we found qualitatively distinct routes to R-tipping, highlighting that the direction and outcome of ecosystem transitions depend strongly

on the underlying structure of plant functional types. In the grassland–forest subsystem, rapid forcing shifts the system toward grass dominance, whereas in the woodland subsystem it favours increased woody cover. In the full system, tipping happens from a coexistence equilibrium to an oscillatory state. These results are broadly consistent with earlier studies showing that savanna–forest systems can exhibit alternative stable states maintained by feedbacks between fire, vegetation structure and climate [35,40,41]. While most of this literature has focused on threshold-driven transitions, our results suggest that rapid change in drivers may provide an additional pathway to trigger such shifts even before the threshold is reached [17,75].

A key challenge in analysing non-autonomous systems is the absence of compact invariant sets, which limits the direct use of classical bifurcation theory even when the ecological processes are well understood. To address this, we reformulated the bi-asymptotically forced savanna–forest model using a compactification approach [65], embedding the original non-autonomous system into an autonomous system with an added bounded dimension. In this setting, R-tipping occurs when trajectories starting from an attractor in the past system fail to track a moving attractor and instead transition toward an alternative stable state in the future system. The critical rate corresponds to the emergence of a heteroclinic connection, offering a clear mathematical characterization of the tipping threshold. Ecologically, this threshold marks the loss of resilience of the original ecosystem state, beyond which recovery becomes unlikely even if environmental conditions stabilize. Overall, this construction provided a geometric interpretation of R-tipping [26].

Despite these insights, our study has certain limitations. The savanna–forest system is represented in an intentionally simplified manner, with four functional types and fire effects incorporated implicitly rather than through explicit spatial dynamics. While key drivers, such as herbivory and land-use change, are captured indirectly through time-dependent parameters, this abstraction allows us to focus on the underlying tipping mechanisms rather than on site-specific complexities. Nonetheless, this perspective is especially relevant in the current context of rapid environmental degradation.

Ongoing and accelerating ecological degradation worldwide, accompanied by widespread biodiversity loss, underscores the need to better understand mechanisms that can trigger abrupt and potentially irreversible change. In this context, greater attention to R-tipping is both timely and necessary, particularly for developing indicators capable of forecasting such transitions and helping to mitigate rate-induced collapse [77–79]. While our study explored possible rate-induced abrupt shifts in savanna–forest ecosystems, an ecosystem type that has received comparatively less attention in the R-tipping literature, it also highlights several directions for future work. In particular, incorporating the effects of fire-mediated feedback through savanna saplings and adult savanna trees ($\gamma > 0$) would allow a more complete representation of fire–vegetation interactions. Since the fire spread is an inherently spatial process, one can further analyse R-tipping spatially using partial differential equation models. Finally, coupling such models with empirical data on fire regimes, vegetation structure and climate variability could help bridge the gap between theoretical advances and real-world ecosystem management.

Data accessibility. MATLAB codes and the necessary data for the figures are available at [80].

The data are provided in electronic supplementary material [81].

Declaration of AI use. We have not used AI-assisted technologies in creating this article.

Authors' contributions. R.K.K.: conceptualization, data curation, formal analysis, investigation, methodology, resources, validation, visualization, writing—original draft, writing—review and editing; P.S.D.: conceptualization, funding acquisition, methodology, project administration, resources, supervision, validation, writing—original draft, writing—review and editing.

Both authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Conflict of interest declaration. We declare we have no competing interests.

Funding. P.S.D. acknowledges financial support from the Science and Engineering Research Board (SERB), Government of India (grant no. CRG/2022/002788). R.K.K. acknowledges IIT Ropar for financial assistance and research facilities.

Appendix A

(a) Conditions on coordinate transformation

The coordinate transformation $s = g_\varepsilon(rt)$ given by equation (3.7) satisfies the following four conditions needed for compactification, where $g: \mathbb{R} \rightarrow (0, 1)$ and $\varepsilon, r > 0$.

1. The sigmoid function g_ε is C^∞ and hence $g_\varepsilon \in C^{k \geq 2}$.
2. As $t \rightarrow +\infty$, $g_\varepsilon(rt) \rightarrow 1$, while as $t \rightarrow -\infty$, $g_\varepsilon(rt) \rightarrow 0$.
3. The derivative $\dot{g}_\varepsilon(rt) = \frac{\varepsilon r e^{-\varepsilon r t}}{(1+e^{-\varepsilon r t})^2} > 0$, $\forall t \in \mathbb{R}$, which ensures that $g(t)$ is a strictly increasing function.
4. The derivative vanishes asymptotically at both infinities: $\lim_{t \rightarrow \pm\infty} \dot{g}_\varepsilon(rt) = 0$.

(b) External inputs under the coordinate transformation

Here, we derive explicitly the inputs Γ_1 equation (3.4) and Γ_2 equation (3.5) under the transformation $s = g_\varepsilon(rt)$ equation (3.7) and its inverse transformation $rt = h_\varepsilon(s) = g_\varepsilon^{-1}(s)$ equation (3.8).

(i) Input 1

$$\begin{aligned} \Gamma_1(rt) &= \Delta_0 + \frac{\Delta_1}{2} \operatorname{sech}(r(t - t_c)) + \Delta_2 \tanh(r(t - t_c)) \\ &= \Delta_0 + \Delta_1 \left(\frac{e^{r(t-t_c)}}{e^{2r(t-t_c)} + 1} \right) + \Delta_2 \left(\frac{e^{2r(t-t_c)} - 1}{e^{2r(t-t_c)} + 1} \right) \\ &= \Delta_0 + \Delta_1 \left(\frac{e^{-rt_c}(s/(1-s))^{1/\varepsilon}}{e^{-2rt_c}(s/(1-s))^{2/\varepsilon} + 1} \right) + \Delta_2 \left(\frac{e^{-2rt_c}(s/(1-s))^{2/\varepsilon} - 1}{e^{-2rt_c}(s/(1-s))^{2/\varepsilon} + 1} \right) \quad (\text{by equation (3.8)}) \end{aligned}$$

and

$$\Gamma_1(h_\varepsilon(s)) = \Delta_0 + \Delta_1 \left(\frac{e^{rt_c} s^{1/\varepsilon} (1-s)^{1/\varepsilon}}{s^{2/\varepsilon} + e^{2rt_c} (1-s)^{2/\varepsilon}} \right) + \Delta_2 \left(\frac{s^{2/\varepsilon} - e^{2rt_c} (1-s)^{2/\varepsilon}}{s^{2/\varepsilon} + e^{2rt_c} (1-s)^{2/\varepsilon}} \right).$$

(ii) Input 2

$$\begin{aligned} \Gamma_2(rt) &= \delta_0 + \delta_1 [\tanh(r(t - t_{c1})) - \tanh(r(t - t_{c2}))] \\ &= \delta_0 + \delta_1 \left(\frac{e^{2r(t-t_{c1})} - 1}{e^{2r(t-t_{c1})} + 1} \right) - \delta_1 \left(\frac{e^{2r(t-t_{c2})} - 1}{e^{2r(t-t_{c2})} + 1} \right) \\ &= \delta_0 + \delta_1 \left(\frac{e^{2rt_{c1}}(s/(1-s))^{2/\varepsilon} - 1}{e^{2rt_{c1}}(s/(1-s))^{2/\varepsilon} + 1} \right) - \delta_1 \left(\frac{e^{2rt_{c2}}(s/(1-s))^{2/\varepsilon} - 1}{e^{2rt_{c2}}(s/(1-s))^{2/\varepsilon} + 1} \right) \quad (\text{by equation (3.8)}) \end{aligned}$$

and

$$\Gamma_2(h_\varepsilon(s)) = \delta_0 + \delta_1 \left(\frac{s^{2/\varepsilon} - e^{2rt_{c1}}(1-s)^{2/\varepsilon}}{s^{2/\varepsilon} + e^{2rt_{c1}}(1-s)^{2/\varepsilon}} \right) - \delta_1 \left(\frac{s^{2/\varepsilon} - e^{2rt_{c2}}(1-s)^{2/\varepsilon}}{s^{2/\varepsilon} + e^{2rt_{c2}}(1-s)^{2/\varepsilon}} \right).$$

(c) Transformation conditions

The compactified system (3.11) is C^1 -smooth on $\mathbb{S}^n \times [0, 1]$ if and only if for $i = 1, 2$,

$$(1) \lim_{t \rightarrow \pm\infty} \frac{\dot{\Gamma}_i(rt)}{\dot{g}_\varepsilon(rt)} \quad \text{and} \quad (2) \lim_{t \rightarrow \pm\infty} \frac{\ddot{g}_\varepsilon(rt)}{\dot{g}_\varepsilon(rt)} \quad \text{exist.}$$

This appendix examines these conditions with the external inputs and the coordinate transformation. The first- and second-order time derivatives of the coordinate transformation

$g_\varepsilon(rt)$ are given by

$$\begin{aligned} \dot{g}_\varepsilon(rt) &= \varepsilon r g_\varepsilon(rt)(1 - g_\varepsilon(rt)) \quad \text{and} \quad \ddot{g}_\varepsilon(rt) = (\varepsilon r)^2 g_\varepsilon(rt)(1 - g_\varepsilon(rt))(1 - 2g_\varepsilon(rt)), \\ (2) \quad \lim_{t \rightarrow \pm\infty} \frac{\ddot{g}_\varepsilon(rt)}{\dot{g}_\varepsilon(rt)} &= \lim_{t \rightarrow \pm\infty} \varepsilon r (1 - 2g_\varepsilon(rt)) \\ &= \lim_{t \rightarrow \pm\infty} \varepsilon r \left(1 - \frac{2}{1 + e^{-\varepsilon rt}} \right) \\ &= \mp \varepsilon r. \quad (\text{finite}) \end{aligned}$$

(i) Input 1

Here, we check the condition (1) for the input Γ_1 equation (3.4). The derivative of $\Gamma_1(rt)$ is

$$\begin{aligned} \dot{\Gamma}_1(rt) &= -\frac{\Delta_1}{2} r \operatorname{sech}(r(t - t_c)) \tanh(r(t - t_c)) + \Delta_2 r \operatorname{sech}^2(r(t - t_c)) \\ &= -\frac{\Delta}{2} r \left(\frac{2}{e^{r(t-t_c)} + e^{-r(t-t_c)}} \right) \left(\frac{e^{r(t-t_c)} - e^{-r(t-t_c)}}{e^{r(t-t_c)} + e^{-r(t-t_c)}} \right) + \Delta_2 r \left(\frac{4}{(e^{r(t-t_c)} + e^{-r(t-t_c)})^2} \right). \end{aligned}$$

For sufficiently large $\pm t$, $\dot{\Gamma}_1(rt) \sim 4r\Delta_2 e^{\mp 2r(t-t_c)} = 4r\Delta_2 e^{\pm 2rt_c} e^{\mp 2rt}$.

The derivative of the coordinate transformation $g_\varepsilon(rt)$ can be written as

$$\dot{g}_\varepsilon(rt) = \frac{\varepsilon r e^{-\varepsilon rt}}{(1 + e^{-\varepsilon rt})^2}.$$

For sufficiently large $\pm t$, $\dot{g}_\varepsilon(rt) \sim \varepsilon r e^{\mp \varepsilon rt}$ and hence

$$\lim_{t \rightarrow \pm\infty} \frac{\dot{\Gamma}_1(rt)}{\dot{g}_\varepsilon(rt)} = \left(\frac{4\Delta_2}{\varepsilon} \right) e^{\pm 2rt_c} \lim_{t \rightarrow \pm\infty} e^{\mp(2-\varepsilon)rt}.$$

The limit exists if the compactification parameter ε lies between 0 and 2, and hence, the compactified system (3.11) with the input Γ_1 equation (3.4) is C^1 -smooth on $\mathbb{S}^n \times [0, 1]$ if and only if $0 < \varepsilon \leq 2$.

(ii) Input 2

Here, we check the condition (1) for the input Γ_2 equation (3.5). The derivative of $\Gamma_2(rt)$ is

$$\begin{aligned} \dot{\Gamma}_2(rt) &= \delta_1 r (\operatorname{sech}^2(r(t - t_{c1})) - \operatorname{sech}^2(r(t - t_{c2}))) \\ &= \delta_1 r \left(\frac{4}{(e^{r(t-t_{c1})} + e^{-r(t-t_{c1})})^2} - \frac{4}{(e^{r(t-t_{c2})} + e^{-r(t-t_{c2})})^2} \right). \end{aligned}$$

For sufficiently large $\pm t$, $\dot{\Gamma}_2(rt) \sim 4r\delta_1 (e^{\mp 2r(t-t_{c1})} - e^{\mp 2r(t-t_{c2})})$ and $\dot{g}_\varepsilon(rt) \sim \varepsilon r e^{\mp \varepsilon rt}$ and hence

$$\lim_{t \rightarrow \pm\infty} \frac{\dot{\Gamma}_2(rt)}{\dot{g}_\varepsilon(rt)} = (4\delta_1/\varepsilon) \left(e^{\pm 2rt_{c1}} \left[\lim_{t \rightarrow \pm\infty} e^{\mp(2-\varepsilon)rt} \right] - e^{\pm 2rt_{c2}} \left[\lim_{t \rightarrow \pm\infty} e^{\mp(2-\varepsilon)rt} \right] \right).$$

The above limits exist if the compactification parameter ε lies between 0 and 2 and hence, the compactified system (3.11) with the input Γ_2 equation (3.5) is C^1 -smooth on $\mathbb{S}^n \times [0, 1]$ if and only if $0 < \varepsilon \leq 2$.

References

1. Flores BM *et al.* 2024 Critical transitions in the amazon forest system. *Nature* **626**, 555–564. (doi:10.1038/s41586-023-06970-0)
2. Van Passel J, Bernardino PN, Lhermitte S, Rius BF, Hirota M, Conradi T, de Keersmaecker W, Van Meerbeek K, Somers B. 2024 Critical slowing down of the amazon forest after increased drought occurrence. *Proc. Natl Acad. Sci. USA* **121**, e2316924121. (doi:10.1073/pnas.2316924121)

3. Ometto J, Kalaba K, Anshari G, Chacón N, Farrell A, Halim S, Neufeldt H, Sukumar R. 2022 Cross- Chapter Paper 7: Tropical Forests, pp. 2369–2410. Cambridge, UK/New York, USA: Cambridge University Press.
4. Lenton TM *et al.* 2025 Global tipping points report 2025.
5. Ravindran S. 2016 Coral reefs at a tipping point. *Proc. Natl Acad. Sci. USA* **113**, 5140–5141. (doi:10.1073/pnas.1605690113)
6. Boers N. 2021 Observation-based early-warning signals for a collapse of the atlantic meridional overturning circulation. *Nat. Clim. Change* **11**, 680–688. (doi:10.1038/s41558-021-01097-4)
7. Boers N, Rypdal M. 2021 Critical slowing down suggests that the western greenland ice sheet is close to a tipping point. *Proc. Natl Acad. Sci. USA* **118**, e2024192118. (doi:10.1073/pnas.2024192118)
8. Ditlevsen P, Ditlevsen S. 2023 Warning of a forthcoming collapse of the atlantic meridional overturning circulation. *Nat. Commun.* **14**, 1–12. (doi:10.1038/s41467-023-39810-w)
9. Morris JT, Sundareshwar P, Nietch CT, Kjerfve B, Cahoon DR. 2002 Responses of coastal wetlands to rising sea level. *Ecology* **83**, 2869–2877. (doi:10.1890/0012-9658(2002)083[2869:ROCWTR]2.0.CO;2)
10. Rotbarth R, van Nes EH, Scheffer M, Holmgren M. 2025 Boreal forests are heading for an open state. *Proc. Natl Acad. Sci. USA* **122**, e2404391121. (doi:10.1073/pnas.2404391121)
11. Ashwin P, Wieczorek S, Vitolo R, Cox P. 2012 Tipping points in open systems: bifurcation, noise-induced and rate-dependent examples in the climate system. *Phil. Trans. R. Soc. A* **370**, 1166–1184. (doi:10.1098/rsta.2011.0306)
12. Thompson JMT, Sieber J. 2011 Predicting climate tipping as a noisy bifurcation: a review. *Int. J. Bifurcation Chaos* **21**, 399–423. (doi:10.1142/S0218127411028519)
13. Cooper GS, Willcock S, Dearing JA. 2020 Regime shifts occur disproportionately faster in larger ecosystems. *Nat. Commun.* **11**, 1175. (doi:10.1038/s41467-020-15029-x)
14. Rocha JC, Peterson GD, Biggs R. 2015 Regime shifts in the anthropocene: drivers, risks, and resilience. *PLoS ONE* **10**, e0134639.
15. Chaparro-Pedraza PC. 2021 Fast environmental change and eco-evolutionary feedbacks can drive regime shifts in ecosystems before tipping points are crossed. *Proc. R. Soc. B* **288**, 20211192. (doi:10.1098/rspb.2021.1192)
16. Hankel C. 2025 The effect of CO₂ ramping rate on the transient weakening of the atlantic meridional overturning circulation. *Proc. Natl Acad. Sci. USA* **122**, e2411357121. (doi:10.1073/pnas.2411357121)
17. Willcock S, Cooper GS, Addy J, Dearing JA. 2023 Earlier collapse of anthropocene ecosystems driven by multiple faster and noisier drivers. *Nature Sustainability* **6**, 1331–1342. (doi:10.1038/s41893-023-01157-x)
18. Kuehn C, Longo IP. 2022 Estimating rate-induced tipping via asymptotic series and a melnikov-like method. *Nonlinearity* **35**, 2559. (doi:10.1088/1361-6544/ac62dc)
19. Longo IP, Núñez C, Obaya R, Rasmussen M. 2021 Rate-induced tipping and saddle-node bifurcation for quadratic differential equations with nonautonomous asymptotic dynamics. *SIAM J. Appl. Dyn. Syst.* **20**, 500–540. (doi:10.1137/20M1339003)
20. Wieczorek S, Ashwin P, Luke CM, Cox PM. 2011 Excitability in ramped systems: the compost-bomb instability. *Proc. R. Soc. A* **467**, 1243–1269. (doi:10.1098/rspa.2010.0485)
21. Alkhayuon HM, Ashwin P. 2018 Rate-induced tipping from periodic attractors: partial tipping and connecting orbits. *Chaos: An Interdiscipl. J. Nonlinear Sci.* **28**, 033608. (doi:10.1063/1.5000418)
22. Feudel U. 2023 Rate-induced tipping in ecosystems and climate: the role of unstable states, basin boundaries and transient dynamics. *Nonlinear Process. Geophys.* **30**, 481–502. (doi:10.5194/npg-30-481-2023)
23. Gil MA, Baskett ML, Munch SB, Hein AM. 2020 Fast behavioral feedbacks make ecosystems sensitive to pace and not just magnitude of anthropogenic environmental change. *Proc. Natl Acad. Sci. USA* **117**, 25 580–25 589. (doi:10.1073/pnas.2003301117)
24. Kumar K R, Alkhayuon H, Wieczorek S, Dutta PS. 2025 Pace in concert with phase: rate-induced phase-tipping in birhythmic oscillators. *Proc. R. Soc. A* **481**, 20240629. (doi:10.1098/rspa.2024.0629)

25. Panahi S, Do Y, Hastings A, Lai Y-C. 2023 Rate-induced tipping in complex high-dimensional ecological networks. *Proc. Natl Acad. Sci. USA* **120**, e2308820120. (doi:10.1073/pnas.2308820120)
26. Wieczorek S, Xie C, Ashwin P. 2023 Rate-induced tipping: thresholds, edge states and connecting orbits. *Nonlinearity* **36**, 3238. (doi:10.1088/1361-6544/acb37)
27. Kaur T, Dutta PS. 2022 Critical rates of climate warming and abrupt collapse of ecosystems. *Proc. R. Soc. A* **478**, 20220086. (doi:10.1098/rspa.2022.0086)
28. Osborne CP, Charles-Dominique T, Stevens N, Bond WJ, Midgley G, Lehmann CE. 2018 Human impacts in african savannas are mediated by plant functional traits. *New Phytol.* **220**, 10–24. (doi:10.1111/nph.15236)
29. Sankaran M. 2019 Droughts and the ecological future of tropical savanna vegetation. *J. Ecol.* **107**, 1531–1549. (doi:10.1111/1365-2745.13195)
30. da Silva Menezes L, Vogel Ely C, Lucas DB, Minervini Silva GH, Boldrini II, Overbeck GE. 2018 Plant species richness record in brazilian pampa grasslands and implications. *Braz. J. Bot.* **41**, 817–823. (doi:10.1007/s40415-018-0492-6)
31. Scholes RJ, Archer SR. 1997 Tree-grass interactions in savannas. *Annu. Rev. Ecol. Syst.* **28**, 517–544. (doi:10.1146/annurev.ecolsys.28.1.517)
32. Dobson A *et al.* 2022 Savannas are vital but overlooked carbon sinks. *Science* **375**, 392–392. (doi:10.1126/science.abn4482)
33. Moustakas A, Kunin WE, Cameron TC, Sankaran M. 2013 Facilitation or competition? Tree effects on grass biomass across a precipitation gradient. *PLoS ONE* **8**, e57025. (doi:10.1371/journal.pone.0057025)
34. Oliveras I, Malhi Y. 2016 Many shades of green: the dynamic tropical forest–savannah transition zones. *Phil. Trans. R. Soc. B* **371**, 20150308. (doi:10.1098/rstb.2015.0308)
35. Staver AC, Levin SA. 2012 Integrating theoretical climate and fire effects on savanna and forest systems. *Am. Nat.* **180**, 211–224. (doi:10.1086/666648)
36. Pilon N *et al.* 2025 Open letter: there are more than just trees and forests to be conserved and restored. *Plants, People, Planet* **7**, 1220–1224. (doi:10.1002/ppp3.10635)
37. Gopalakrishna T, Rifai SW, Ratnam J, Oliveras Menor I, Stevens N, Malhi Y. 2024 The distribution and drivers of tree cover in savannas and forests across india. *Commun. Earth Environ.* **5**, 399. (doi:10.1038/s43247-024-01568-4)
38. Fernandes GW, Coelho MS, Machado RB, Ferreira ME, de Souza Aguiar LM, Dirzo R, Scariot A, Lopes. Afforestation of savannas CR. 2016 an impending ecological disaster. *Natureza & Conservação* **14**, 146–151. (doi:10.1016/j.ncon.2016.08.002)
39. Veldman JW, Overbeck GE, Negreiros D, Mahy G, Le Stradic S, Fernandes GW, Durigan G, Buisson E, Putz FE, Bond WJ. 2015 Where tree planting and forest expansion are bad for biodiversity and ecosystem services. *BioScience* **65**, 1011–1018. (doi:10.1093/biosci/biv118)
40. Hirota M, Holmgren M, Van Nes EH, Scheffer M. 2011 Global resilience of tropical forest and savanna to critical transitions. *Science* **334**, 232–235. (doi:10.1126/science.1210657)
41. Staver AC, Archibald S, Levin S. 2011 Tree cover in sub-saharan africa: rainfall and fire constrain forest and savanna as alternative stable states. *Ecology* **92**, 1063–1072. (doi:10.1890/10-1684.1)
42. Dohn J, Dembélé F, Karembé M, Moustakas A, Amévor KA, Hanan NP. 2013 Tree effects on grass growth in savannas: competition, facilitation and the stress-gradient hypothesis. *J. Ecol.* **101**, 202–209. (doi:10.1111/1365-2745.12010)
43. Archibald S, Roy DP, van Wilgen BW, Scholes RJ. 2009 What limits fire? An examination of drivers of burnt area in Southern Africa. *Glob. Change Biol.* **15**, 613–630. (doi:10.1111/j.1365-2486.2008.01754.x)
44. Grady JM, Hoffmann WA. 2012 Caught in a fire trap: recurring fire creates stable size equilibria in woody resprouters. *Ecology* **93**, 2052–2060. (doi:10.1890/12-0354.1)
45. Hanan NP, Sea WB, Dangelmayr G, Govender N. 2008 Do fires in savannas consume woody biomass? A comment on approaches to modeling savanna dynamics. *Am. Nat.* **171**, 851–856. (doi:10.1086/587527)
46. Higgins SI, Bond WJ, Trollope WS. 2000 Fire, resprouting and variability: a recipe for grass–tree coexistence in savanna. *J. Ecol.* **88**, 213–229. (doi:10.1046/j.1365-2745.2000.00435.x)
47. Hoffmann WA, Orthen B, Do Nascimento PKV. 2003 Comparative fire ecology of tropical savanna and forest trees. *Funct. Ecol.* **17**, 720–726. (doi:10.1111/j.1365-2435.2003.00796.x)
48. Schertzer E, Staver A, Levin SA. 2015 Implications of the spatial dynamics of fire spread for the bistability of savanna and forest. *J. Math. Biol.* **70**, 329–341. (doi:10.1007/s00285-014-0757-z)

49. Brando PM, Nepstad DC, Balch JK, Bolker B, Christman MC, Coe M, Putz FE. 2012 Fire-induced tree mortality in a neotropical forest: the roles of bark traits, tree size, wood density and fire behavior. *Glob. Change Biol.* **18**, 630–641. (doi:10.1111/j.1365-2486.2011.02533.x)
50. Van Langevelde F *et al.* 2003 Effects of fire and herbivory on the stability of savanna ecosystems. *Ecology* **84**, 337–350. (doi:10.1890/0012-9658(2003)084[0337:EOFAHO]2.0.CO;2)
51. Nerlekar AN, Munje A, Mhaisalkar P, Hiremath AJ, Veldman JW. 2024 Tillage agriculture and afforestation threaten tropical savanna plant communities across a broad rainfall gradient in India. *J. Ecol.* **112**, 98–109. (doi:10.1111/1365-2745.14221)
52. Honda EA, Durigan G. 2016 Woody encroachment and its consequences on hydrological processes in the savannah. *Phil. Trans. R. Soc. B* **371**, 20150313. (doi:10.1098/rstb.2015.0313)
53. Pellegrini AF *et al.* 2018 Fire frequency drives decadal changes in soil carbon and nitrogen and ecosystem productivity. *Nature* **553**, 194–198. (doi:10.1038/nature24668)
54. Touboul JD, Staver AC, Levin SA. 2018 On the complex dynamics of savanna landscapes. *Proc. Natl Acad. Sci. USA* **115**, E1336–E1345. (doi:10.1073/pnas.1712356115)
55. Ratajczak Z, D'Odorico P, Collins SL, Bestelmeyer BT, Isbell FI, Nippert JB. 2017 The interactive effects of press/pulse intensity and duration on regime shifts at multiple scales. *Ecol. Monogr.* **87**, 198–218. (doi:10.1002/ecm.1249)
56. Scheffer M, Van Nes EH, Holmgren M, Hughes T. 2008 Pulse-driven loss of top-down control: the critical-rate hypothesis. *Ecosystems* **11**, 226–237. (doi:10.1007/s10021-007-9118-8)
57. Staal A, van Nes EH, Hantson S, Holmgren M, Dekker SC, Pueyo S, Xu C, Scheffer M. 2018 Resilience of tropical tree cover: the roles of climate, fire, and herbivory. *Glob. Change Biol.* **24**, 5096–5109. (doi:10.1111/gcb.14408)
58. O'Sullivan E, Mulchrone K, Wieczorek S. 2023 Rate-induced tipping to metastable zombie fires. *Proc. R. Soc. A* **479**, 20220647. (doi:10.1098/rspa.2022.0647)
59. Alkhayun H, Tyson RC, Wieczorek S. 2021 Phase tipping: how cyclic ecosystems respond to contemporary climate. *Proc. R. Soc. A* **477**, 20210059. (doi:10.1098/rspa.2021.0059)
60. O'Keeffe PE, Wieczorek S. 2020 Tipping phenomena and points of no return in ecosystems: beyond classical bifurcations. *SIAM J. Appl. Dyn. Syst.* **19**, 2371–2402. (doi:10.1137/19M1242884)
61. Ritchie PD, Alkhayun H, Cox PM, Wieczorek S. 2023 Rate-induced tipping in natural and human systems. *Earth Syst. Dyn.* **14**, 669–683. (doi:10.5194/esd-14-669-2023)
62. Ashwin P, Perryman C, Wieczorek S. 2017 Parameter shifts for nonautonomous systems in low dimension: bifurcation and rate-induced tipping. *Nonlinearity* **30**, 2185. (doi:10.1088/1361-6544/aa675b)
63. Longo IP, Núñez C, Obaya R. 2024 Critical transitions in piecewise uniformly continuous concave quadratic ordinary differential equations. *J. Dyn. Differ. Equ.* **36**, 2153–2192. (doi:10.1007/s10884-022-10225-3)
64. Hasan CR, Cárthaigh RM, Wieczorek S. 2023 Rate-induced tipping in heterogeneous reaction-diffusion systems: An invariant manifold framework and geographically shifting ecosystems. *SIAM J. Appl. Dyn. Syst.* **22**, 2991–3024. (doi:10.1137/22M1536625)
65. Wieczorek S, Xie C, Jones CK. 2021 Compactification for asymptotically autonomous dynamical systems: theory, applications and invariant manifolds. *Nonlinearity* **34**, 2970. (doi:10.1088/1361-6544/abe456)
66. Lawes MJ, Adie H, Russell-Smith J, Murphy B, Midgley JJ. 2011 How do small savanna trees avoid stem mortality by fire? The roles of stem diameter, height and bark thickness. *Ecosphere* **2**, 1–13. (doi:10.1890/ES10-00204.1)
67. Hoyer-Leitzel A, Iams S. 2021 Impulsive fire disturbance in a savanna model: Tree–grass coexistence states, multiple stable system states, and resilience. *Bull. Math. Biol.* **83**, 113. (doi:10.1007/s11538-021-00944-x)
68. Li B-L. 2002 A theoretical framework of ecological phase transitions for characterizing tree-grass dynamics. *Acta Biotheor.* **50**, 141–154. (doi:10.1023/A:1016562208170)
69. Sullivan AL. 2009 Wildland surface fire spread modelling, 1990–2007. 3: simulation and mathematical analogue models. *Int. J. Wildland Fire* **18**, 387–403. (doi:10.1071/WF06144)
70. Xie C. 2020 Rate-induced critical transitions. PhD thesis, University College Cork.
71. Kloeden PE, Rasmussen M. 2011 *Nonautonomous dynamical systems*, vol. 176. Providence, Rhode Island: American Mathematical Society.
72. Hoffmann WA, Geiger EL, Gotsch SG, Rossatto DR, Silva LCR, Lau OL, Haridasan M, Franco AC. 2012 Ecological thresholds at the savanna–forest boundary: how plant traits,

- resources and fire govern the distribution of tropical biomes. *Ecol. Lett.* **15**, 759–768. (doi:10.1111/j.1461-0248.2012.01789.x)
73. Bond WJ, Midgley JJ. 2001 Ecology of sprouting in woody plants: the persistence niche. *Trends Ecol. Evol.* **16**, 45–51. (doi:10.1016/S0169-5347(00)02033-4)
 74. McKay DIA *et al.* 2023 Tipping points in the biosphere.
 75. Abbott KC, Heggerud CM, Lai Y-C, Morozov A, Petrovskii S, Cuddington K, Hastings A. 2024 When and why ecological systems respond to the rate rather than the magnitude of environmental changes. *Biol. Conservat.* **292**, 110494. (doi:10.1016/j.biocon.2024.110494)
 76. Synodinos AD *et al.* 2023 The rate of environmental change as an important driver across scales in ecology. *Oikos* **2023**, e09616. (doi:10.1111/oik.09616)
 77. Deb S, Dutta PS. 2024 Deep learning at the forefront of detecting tipping points. *Nature Mach. Intell.* **6**, 1433–1434. (doi:10.1038/s42256-024-00957-w)
 78. Huang Y, Bathiany S, Ashwin P, Boers N. 2024 Deep learning for predicting rate-induced tipping. *Nature Mach. Intell.* **6**, 1556–1565. (doi:10.1038/s42256-024-00937-0)
 79. Ritchie P, Sieber J. 2016 Early-warning indicators for rate-induced tipping. *Chaos: An Interdiscipl. J. Nonlinear Sci.* **26**, 093116. (doi:10.1063/1.4963012)
 80. Ravi Kumar K, Dutta PS. 2026 Rate-induced tipping in savanna–forest ecosystems: effects of fast-varying fitness parameters in the framework of compactification. Zenodo. (doi:10.5281/zenodo.18005792)
 81. Ravi Kumar K, Dutta PS. 2026 Rate-induced tipping in savanna–forest ecosystems: effects of fast-varying fitness parameters in the framework of compactification. Figshare. (doi:10.6084/m9.figshare.c.8407476)